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Owner-Side. Vendor-Neutral. Evidence-Led.

THE SAP VALUE PROTECTION GUIDE

What CEOs, CFOs, CIOs & CSCOs Should Prove Before Buying More SAP

S/4HANA • RISE • Ariba • EAM • BTP • Joule • AI • Licensing • Clean Core • Master Data

A practical executive guide for Energy, Oil & Gas, Power, Utilities, EPC and Heavy-Asset Industries

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Buy what you need.

Prove what you need before you buy it.

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BEFORE YOU READ

Evidence Note and How to Use This Guide

This guide is written for the people who sign SAP contracts, approve SI statements of work and answer to a board when the value does not arrive.

> **NOTE | Disclaimer and evidence note**
> This guide is an independent client-side advisory framework. SAP product capabilities, licensing, contractual terms and availability can vary by release, edition, geography, commercial agreement and customer configuration. Readers should validate commercial and technical conclusions against current SAP documentation and their own contractual documents before making purchasing or architecture decisions.

Why this guide exists

Heavy-asset companies - refineries, LNG trains, offshore platforms, power stations, grids, terminals, mines and the EPC contractors who build them - are among the most complex SAP estates in the world. They carry decades of configuration, thousands of custom objects, safety-critical maintenance processes and procurement volumes measured in billions. They are also the estates where a single weak decision - an unnecessary product, an unchallenged work package, an over-sized licence commitment - compounds for seven years or more.

This guide does not argue that SAP is unnecessary. SAP is a powerful enterprise platform and, for most of the organisations reading this, it will remain the system of record for finance, supply chain and asset management. The argument is narrower and more demanding: **buy what you genuinely need, but prove what you need before you buy it.** Every licence, add-on, cloud migration, AI subscription, integration and line of custom code should earn its place with evidence.

How the evidence is labelled

The primary source for this guide was an extended advisory working document that explored software economics, SAP deployment models, licensing, AI agents, clean core and owner-side governance. That document contained valuable concepts, but it also contained claims that were interpretations, outdated, or not verifiable. Rather than filling the guide with generic disclaimers, each important statement is labelled where it appears:

Label	What it means	How to use it
VERIFIED FACT	Confirmed against SAP documentation, SAP announcements or credible reporting listed in the Evidence Register (Appendix B). Source references appear in square brackets, for example [S03].	Safe to use in a steering committee, but still check the date - SAP changes pricing and policy frequently.
SOURCE-DOCUMENT POSITION	A view taken in the source working document that is useful but has not been independently established.	Treat as a hypothesis to test, not a fact to quote.
ADVISORY INTERPRETATION	The JM ETA reading of the facts from the owner side of the table.	Use as a challenge position; ask the vendor to disprove it with evidence.
CLIENT-SPECIFIC VALIDATION REQUIRED	The answer depends on your release, edition, contract, order form, geography or configuration.	Do not assume. Obtain the clause, the document or the written confirmation.
CURRENT FACT / EMERGING TREND / STRATEGIC SCENARIO	Used in the AI chapters to separate what is true today from what is developing and what might happen.	Never let a scenario be presented to your board as a certainty.

Label	What it means	How to use it
ILLUSTRATIVE ONLY	Arithmetic or examples built to explain a mechanism. No real prices or customer data.	Replace with your own contract numbers before any decision.

Who should use it and how

The guide is organised for executives who have limited time and a large decision in front of them. Part I explains the economics that are shifting under enterprise software. Part II addresses the architecture decisions - RISE, deployment models, upgrades, custom code, BTP and Ariba. Part III addresses the foundation that determines whether any of it works: asset data, master data, procurement governance and data ownership. Part IV is the toolkit - the challenge framework, the CEO 100 questions, the red-flag statements, the value protection checklist and the golden rules - designed to be printed and taken into a steering committee.

Nothing in this guide is a substitute for reading your own contract. It is a structured way to ask the questions that the contract must answer.

FOREWORD

The One Question Before the Next Dollar

Before we spend another dollar on SAP, an SI, an add-on, a licence, a cloud migration, an AI product or custom development - what should we prove first?

Most SAP value leakage in heavy industry does not come from a single catastrophic decision. It comes from a sequence of reasonable-sounding approvals. A licence bought "for flexibility". An add-on bought because "everyone in the industry uses it". A custom development approved because "SAP cannot do that". A migration scope accepted because "this is the recommended architecture". An AI subscription signed because "you will be left behind". Each approval is small relative to the programme. Together they define a cost base that the owner carries for the life of the contract.

The purpose of this guide is to put one discipline in front of every one of those approvals: **prove it first.** Prove the requirement. Prove that standard SAP cannot meet it. Prove the licence you actually need. Prove that the architecture is sound for a heavy-asset operation. Prove the value over five and seven years. Then spend.

Ten messages for the executive committee

1. **SAP is not the problem; unproven purchasing is.** The objective is not to buy less SAP for the sake of it. It is to buy exactly what the business needs and to challenge everything else with evidence.
2. **"SAP cannot do it" is a claim, not a finding.** It must be supported by a requirement definition, a standard-functionality assessment, a release check and a licence check before it can justify spend.
3. **The economics of enterprise software are shifting.** SAP has publicly signalled a move towards consumption-based pricing for AI, while core cloud ERP remains priced largely on user-access equivalents. Contracts signed today must survive tomorrow's automation.
4. **AI agents change who operates the ERP, not whether you need one.** The system of record, its rules and its audit trail become more important, not less, when machines execute transactions.
5. **RISE is a commercial and operating model, not a transformation.** It moves infrastructure and technical operations to SAP; it does not clean your data, design your processes or remediate your custom code.
6. **There is no universal winner between Public, Private and On-Premise.** Each can be right. The decision belongs to evidence about scope, control, data rules and total cost, not to a sales roadmap.
7. **Customisation is not automatically technical debt.** Uncontrolled customisation is. SAP's own clean core guidance distinguishes levels of extension quality rather than banning custom logic.
8. **Ariba can run with On-Premise S/4HANA or ECC.** A cloud procurement front-end does not require a cloud ERP. It does require clean material, supplier and asset data on the owner side.
9. **AI cannot rescue a broken asset foundation.** Functional locations, equipment, BOMs, material masters, MPNs and failure history decide whether automation creates value or accelerates error.
10. **The faster the machine, the stronger the governance must be.** Delegation of Authority, segregation of duties, price and contract validation and audit evidence must be designed for agents before agents are switched on.

JM ETA ADVISORY PRINCIPLE | The central warning

Never approve a new SAP product, licence, add-on, custom development or major SI work package merely because someone says "SAP cannot do it."

Require the requirement definition, the standard-functionality assessment, the configuration assessment, the release and version assessment, the licence assessment, the architecture assessment, the clean-core assessment, the TCO assessment, the alternatives assessment - and the documents that prove each one.

PART I

The Economics Are Shifting

Software deflation, AI agents and the two-meter licensing problem - what is fact, what is trend and what is still a scenario.

CHAPTER 1

The JM ETA Governance Sequence

PROVE → STANDARD → CONFIGURE → EXTEND → INTEGRATE → CUSTOMISE → BUY. Every step to the right adds cost, dependency and lock-in. Nothing moves right without evidence.

Every SAP decision in this guide is tested against one sequence. It is deliberately ordered from the cheapest, least dependent and most reversible option to the most expensive, most dependent and least reversible one. The sequence does not forbid any option - including buying a new product - but it demands that each earlier option has been genuinely evaluated and documented before the next is approved.

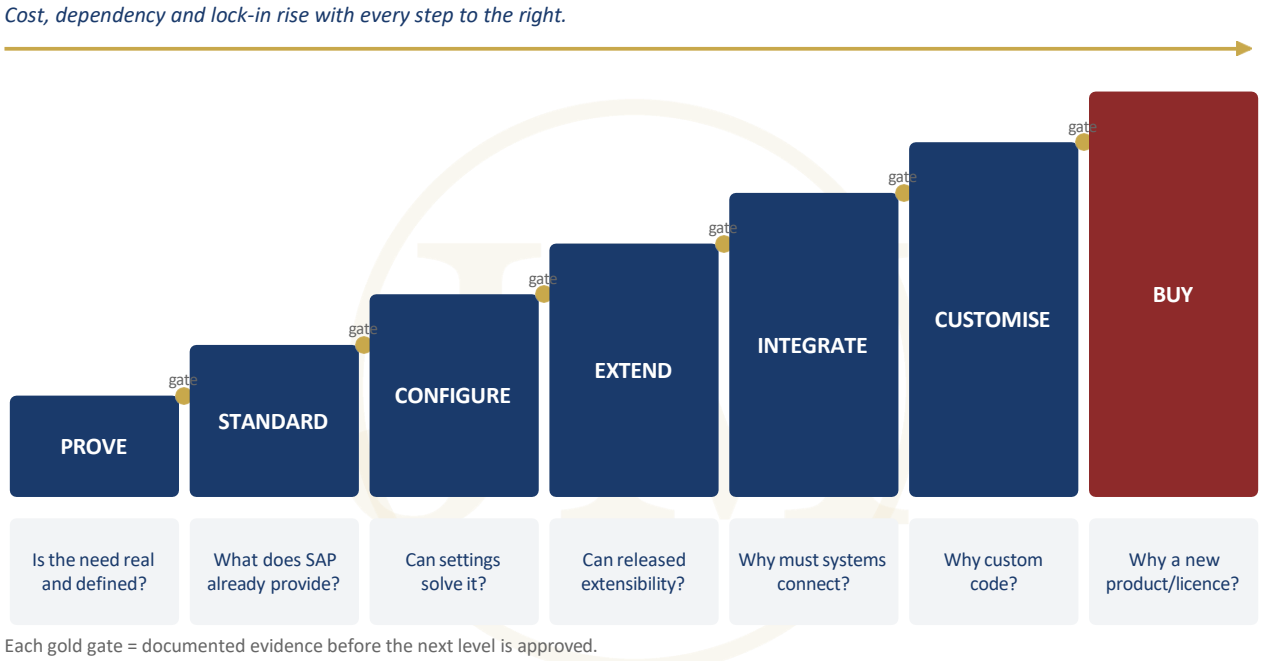


Figure. The JM ETA governance sequence. Buying a new product or licence is the last step, not the first.

The seven levels explained

1. PROVE - the requirement

Before any solution is discussed, the business requirement must be written in business terms: what outcome, for which process, at what volume, with what control and what measurable benefit. Many expensive SAP decisions fail at this step because the "requirement" was written by the party proposing the solution. A requirement that names a product is not a requirement; it is a purchase order in disguise.

2. STANDARD - what SAP already provides

The first technical question is always: what does standard SAP already provide for this process in our release and edition? The second is: what do we already own - in licences, subscriptions, BTP entitlements, industry solutions and previously purchased add-ons that are sitting unused? Heavy-asset estates frequently discover that functionality being proposed as new was licensed years ago and never activated.

3. CONFIGURE - can settings solve it?

Configuration is the cheapest form of change and the easiest to upgrade. Release strategies, workflow, output management, classification, field status, user exits that SAP intends for use, and business rules often solve requirements that are initially presented as development. If the answer is "configuration cannot do it", the owner should see which configuration options were evaluated and why each failed.

4. EXTEND - supported, upgrade-stable extensibility

Where configuration is not enough, supported extensibility - key-user extensibility, developer extensibility using released interfaces, or side-by-side extensions - should be evaluated next. SAP's current clean core guidance grades extensions by upgrade stability rather than by location, and this matters for heavy industry: a well-built extension inside the core can be cleaner than a poorly designed one outside it (see Chapter 8).

5. INTEGRATE - why must systems connect?

Every integration is a permanent operating cost: monitoring, error handling, security, testing at every upgrade and a support owner who answers when a message fails at 3 a.m. during a turnaround. Integration should be approved because data genuinely needs to move between systems of record, not because a new product has been bought that now needs feeding.

6. CUSTOMISE - why custom code?

Custom development is legitimate when it encodes genuine business differentiation or regulatory obligation that standard and supported extensibility cannot meet. It must be owned, documented, tested and periodically reviewed for retirement. The question is never "is custom code bad?" but "is this custom code justified, governed and the least-dependent way to meet the need?"

7. BUY - why another product or licence?

A new SAP product, add-on or licence type is the most dependent decision of all. It usually brings a subscription that renews, an integration that must be maintained, a skills requirement and a new negotiation at every renewal. It should be approved only when the previous six levels have been evaluated and documented, and when the five- and seven-year total cost of ownership has been modelled against the alternatives.

The question ladder

In a steering committee, the sequence becomes a ladder of questions. Ask them in order and do not move to the next until the current one has a documented answer:

1. What does SAP already provide?
2. What do we already own?
3. Can standard configuration satisfy the requirement?
4. Can supported clean-core extensibility satisfy it?
5. Why is integration required?
6. Why is custom development required?
7. Why is another SAP product required?
8. What exactly are we paying for - which licence metric, which quantity, for how long, with which exit?

EVIDENCE REQUIRED | What to demand before approval

- Requirement definition written in business terms, owned by the process owner.
- Standard-functionality assessment for your release and edition, with SAP documentation references.
- Configuration assessment showing options tried and why they failed.
- Release and version assessment - is the capability in a later feature pack or release you are already entitled to?
- Licence assessment - is this covered by what you already own?
- Architecture and clean-core assessment - where will this live and how will it survive upgrades?
- TCO assessment over five and seven years, including integration, testing and support.
- Alternatives assessment - including "do nothing" and "fix the data first".

CHAPTER 2

The Great Software Deflation

For a decade, enterprise software behaved like a tax on headcount. The question now is whether today's licensing commitments are flexible enough for tomorrow's automation economics.

What the source document argues

SOURCE-DOCUMENT POSITION The source working document opened with a thesis circulating widely among technology executives: enterprise software has operated as a tax on headcount; AI agents can now operate enterprise software the way humans do, at a very different speed and cost; and therefore the seat-based pricing model is becoming misaligned with how work is executed. It described three shifts - the end of the seat-based model, the move from product to infrastructure, and the rise of a "sovereign data moat" - and concluded that the executive question is no longer "which software should we buy?" but "why are we still paying for the interface when we only need the output?"

This is a powerful framing. It is also a framing that can be misused - by those who want to argue that ERP is finished, and by those who want to sell a new AI layer on top of it. The owner-side task is to separate what has actually happened from what might happen.

What can be verified today

VERIFIED FACT An analysis reported by The Information, using the AlphaSense research platform, found that 27 software companies - including Figma, Workday and HubSpot - listed AI agents as a competitive risk in their securities filings in early 2026, compared with seven companies in the same period a year earlier. [S23]

VERIFIED FACT AI developers have released agents capable of operating business applications through the same interfaces humans use. Reporting in February 2026 described a preview agent from Anthropic that can take over a user's mouse and keyboard to work across applications, and an OpenAI tool designed to help businesses run large numbers of agents across software applications. [S24]

VERIFIED FACT SAP's chief executive stated in March 2026 that, as customers adopt SAP's AI tools, SAP will charge based on AI consumption, moving away from conventional subscription pricing. [S25]

VERIFIED FACT SAP publishes that its SAP-delivered autonomous enterprise agents in 2026 are priced at 0.02 AI Units per agent action, where an action is a single step such as calling a service or triggering a workflow, and that Base AI is included in SAP cloud subscriptions without consuming AI Units. [S01]

What cannot be stated as fact

SOURCE-DOCUMENT POSITION The claims that "software valuations are adjusting because headcount-linked revenue is becoming a declining model" and that "the seat-based model has ended" are interpretations. Share prices move for many reasons, and SAP's core cloud ERP subscriptions remain priced on Full Use Equivalents, a weighted measure of human user access (Chapter 4).

ADVISORY INTERPRETATION The accurate statement for an executive committee is narrower and more useful: **the pricing of the AI layer is moving towards consumption, while the pricing of the ERP core remains largely tied to human access.** Both meters will exist in many contracts for years. The risk for the owner is not that one model replaces the other overnight; it is that a long commitment on one meter is signed without modelling how the other will grow.

Separating fact, trend and scenario

Statement	Classification	Owner-side implication
AI agents can operate business applications through user interfaces and APIs.	CURRENT FACT [S24]	Automation of screen-based work is no longer theoretical. Pilot it, measure it, govern it.
Software vendors are formally disclosing AI agents as a competitive risk.	CURRENT FACT [S23]	Vendors are reshaping pricing and access rules to protect revenue. Expect commercial change.
SAP is pricing premium AI and agents by consumption (AI Units).	CURRENT FACT [S01]	A second meter now exists. It needs a forecast, an owner and a cap.
Human user counts in ERP will fall materially as agents absorb transactional work.	EMERGING TREND	Model the effect on FUE commitments before signing multi-year terms.
ERP will become a "silent database" operated almost entirely by agents.	STRATEGIC SCENARIO	Plausible for some processes; unproven for safety-critical and judgement-heavy ones. Do not plan the enterprise on it.
Seat-based licensing will disappear.	STRATEGIC SCENARIO	Do not assume it. Negotiate as though both meters will coexist for the full term.

> **CEO QUESTION | The CFO question for every renewal**
>
> "Are today's licensing commitments sufficiently flexible for tomorrow's automation economics?"
>
> Ask the vendor and your SI to show, in writing, what happens to your committed quantities if automation removes 10%, 25% and 40% of transactional user activity during the term. If nobody can answer, the contract has not been modelled - it has only been priced.

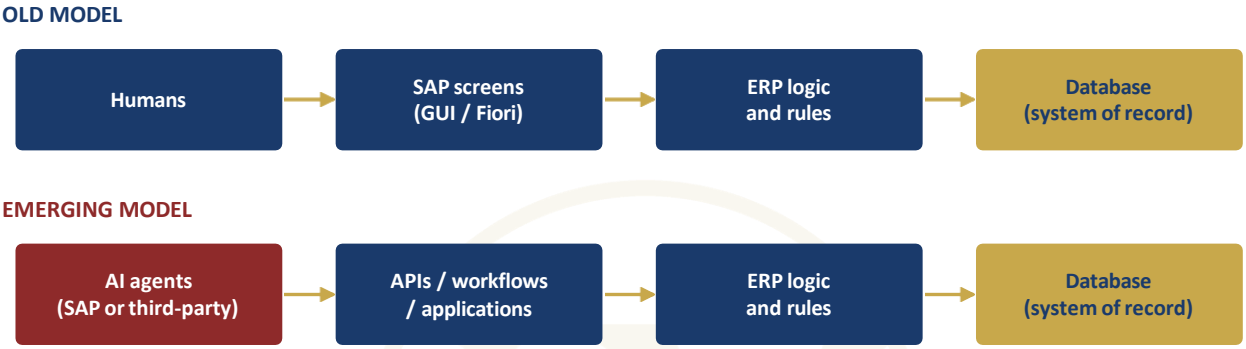
Software deflation, properly understood, is not a prediction that software becomes cheap. It is a recognition that the unit of value is shifting - from the human who clicks, to the transaction that completes, to the decision that is made on trusted data. Owners who negotiate on the old unit while their operations move to the new one will pay twice.

CHAPTER 3

From Screens to Agents

AI agents do not eliminate the need for an ERP's system of record merely because they may reduce the number of humans who use its screens.

The shift in operating model



What changes: who (or what) interacts, and through which interface.
What does not change: the need for one governed, auditable record of truth.

Figure. The old and emerging interaction models. The interface changes; the need for a governed record of truth does not.

For thirty years the ERP operating model has been simple: people log on to SAP screens, transactions pass through ERP logic, and the database records the result. Licences, training, authorisations and support were all designed around that human interaction. The emerging model inserts a new actor. AI agents - SAP-delivered or built by the owner - receive a request, reason about it, call APIs or workflows, and let the ERP record the outcome.

ADVISORY INTERPRETATION The source working document suggested that agentic AI will "hollow out" ERP until it becomes a silent database. The more precise owner-side reading is that the ERP's role shifts from being the place where people work to being the place where rules are enforced and truth is recorded. The ledger, the asset register, the inventory position, the approval history and the audit trail become more valuable when machines execute at speed, because they are the only defence against automated error.

What changes across the enterprise

The following table sets out the likely consequences by domain. It separates what is true today from what is developing and what remains a scenario, so that no board paper confuses them.

Domain	Current fact	Emerging trend	Strategic scenario
SAP user populations	Cloud ERP access is licensed by weighted user types (FUE) [S03].	Transactional roles shrink as agents absorb routine entry and matching.	Some processes run with very few named users.
FUE commitments	Committed for the subscription term; reductions typically addressed at renewal [S04].	Owners seek reclassification and conversion rights.	Access-based metrics weaken for highly automated processes.
Software consumption	Premium AI consumes AI Units [S01].	Consumption meters spread across AI services.	Outcome-based pricing for defined processes.

Domain	Current fact	Emerging trend	Strategic scenario
AI Units	Purchased annually; unused units expire after 12 months [S01].	Forecasting and caps become finance disciplines.	AI Units become a material budget line.
Automation	Agents can operate applications through UIs and APIs [S24].	Agent orchestration across SAP and non-SAP.	Straight-through processing for defined categories.
Workforce	Roles remain accountable for approvals and exceptions.	Shift from data entry to exception management and oversight.	Smaller transactional teams; larger governance roles.
Procurement	DoA and SoD are enforced through ERP workflow and policy.	Agents draft requisitions, check contracts, prepare POs.	Autonomous buying within tight categories and limits.
EAM	Work management depends on FL, equipment, BOM and history.	Agents triage notifications and propose job plans.	Condition-driven autonomous planning where data is mature.
Inventory	Stock accuracy drives availability of critical spares.	Agents propose reorder and reservation actions.	Automated replenishment for governed materials.
Finance	Postings must remain auditable and compliant.	Automated matching and reconciliation expand.	Continuous close in highly standardised entities.
Master data	Quality determines every downstream result.	Agents propose data corrections for steward approval.	Data governance becomes the core operating control.
ERP interfaces	Fiori/GUI remain primary for humans.	Conversational interfaces (e.g. Joule) grow.	Agents become the main "user" for routine work.
APIs	Access is governed by SAP API policy and contract [S09].	Vendors define endorsed agent pathways.	API access becomes a commercial battleground.
Governance	Controls designed around human actors.	Agent identities, limits and logging are added.	Agent governance becomes a board-level control.
Auditability	Auditors test human approvals and system controls.	Auditors ask for agent decision logs.	Evidence of agent reasoning becomes mandatory.

Two routes to automation - and the new policy reality

The source document identified two routes. In the first, the owner buys SAP-delivered AI assistants and agents; human effort falls, and the vendor recovers revenue through AI Units. In the second, the owner builds or buys third-party agents that interact with SAP through APIs; human effort falls, and the owner keeps more of the saving. The document concluded that the second route "captures 100% of the financial upside". That conclusion now needs to be tested against two developments.

VERIFIED FACT In April 2026 SAP published version 4 of its API Policy. Section 2.2.2 restricts use of SAP APIs for interaction or integration with semi-autonomous or generative AI systems that plan, select or execute sequences of API calls, except through SAP-endorsed architectures. Independent analysts reported that technical enforcement for certain extraction methods began on 9 June 2026, and SAP's chief executive said publicly that the intent was to protect SAP's domain know-how and platform stability rather than to block customers from their own data. [S09, S10, S11]

VERIFIED FACT SAP's original position, stated in 2023, was that its newest innovations would be delivered only in the cloud through RISE. At SAP Sapphire in May 2026, SAP said that the majority of Joule assistants and agents would also be made available for on-premise S/4HANA and ECC customers who have committed the majority of their landscape to the RISE journey, as an interim arrangement tied to a specific commercial plan. [S05]

CLIENT-SPECIFIC VALIDATION REQUIRED Whether the API Policy applies to your contract - particularly to perpetual on-premise licences signed before 2026 - whether your intended agent architecture is "endorsed", and whether you qualify for SAP AI on on-premise systems are contractual questions. Obtain the policy version referenced by your agreement and a written position from SAP before designing a third-party agent architecture around SAP data.

ADVISORY INTERPRETATION The practical conclusion is that "bring your own AI" is not automatically cheaper or freer, and "buy SAP AI" is not automatically safer or more integrated. Both must be modelled: licence and consumption cost, API and policy compliance, data movement, security, support and exit. The architecture that looks cheapest on a slide may be the one that breaches a policy or cannot be supported at the next upgrade.

RED FLAG | "You have to move to Public Cloud to use AI."

This was never a complete statement, and it is less accurate now. Joule has been available in S/4HANA Cloud Private Edition from release 2021 [S08], and SAP has announced AI availability for eligible on-premise customers on specific commercial terms [S05].

What to ask next: which specific assistants and agents are available for our edition and release, what are the prerequisites (BTP, identity services, UI technology), which licence or plan is required, and what does each consume in AI Units?

CHAPTER 4

Licensing Under the Microscope

Two meters now run in many SAP cloud contracts: one counts human access, the other counts machine action. The owner must understand both before committing to either.

Meter one: Full Use Equivalents

VERIFIED FACT Under RISE with SAP, S/4HANA Cloud access is licensed in Full Use Equivalents. SAP describes the ratios as 1 FUE for one advanced-use user, 1 FUE for five core-use users and 1 FUE for thirty self-service users, with developer access weighted at 2 FUE per developer. [S03]

SOURCE-DOCUMENT POSITION The source document described FUE as "a disguised user count". That is a fair description of its economic behaviour: FUE converts human access into a single pooled unit. It is also more flexible than the named-user model it replaced, because the pool can be reallocated between user types during the term.

VERIFIED FACT Independent licensing advisers report that SAP does not publish a list price per FUE, that the ratios themselves do not change in negotiation, and that committed FUE quantities are not reduced mid-term - the opportunity to reduce comes at renewal. [S04]

CLIENT-SPECIFIC VALIDATION REQUIRED Your order form, not a market report, determines whether you have reclassification rights, reduction rights, true-up rules, renewal caps or conversion rights. Read those clauses before modelling any automation saving.

The insight most boards miss: automation savings depend on which users disappear

Because FUE weights users by access level, the licence effect of automation is not proportional to the number of people whose work is automated. It depends on their classification.

ILLUSTRATIVE ONLY Consider an estate where agents absorb the work of 300 self-service users, 250 core users and 20 advanced users. In FUE terms, the 300 self-service users represent 10 FUE, the 250 core users represent 50 FUE and the 20 advanced users represent 20 FUE. The headline "we automated 570 users" translates into 80 FUE of potential reduction - and only if the contract permits reduction and the remaining users are correctly classified. No prices are implied; the arithmetic uses SAP's published ratios only [S03].

The same logic works in reverse. A single unnecessary authorisation can promote a user from core to advanced classification. Independent advisers report that over-classification materially inflates FUE totals in many estates [S04]. Before any RISE negotiation, the owner should run its own user-classification analysis and challenge every advanced user.

Meter two: AI Units

VERIFIED FACT SAP describes AI Units as a consumption-based currency purchased annually to activate premium AI capabilities across SAP cloud services; unused AI Units expire after 12 months. Base AI, including Joule navigational scenarios and simple transactional capabilities, is included in SAP cloud subscriptions at no additional cost. Premium AI is priced per user per month or by consumption depending on the feature. SAP-delivered autonomous agents are priced at 0.02 AI Units per agent action in 2026. [S01]

VERIFIED FACT SAP Joule for Consultants is charged at 35 AI Units per assigned user per month, including a pooled request allowance. The source working document attributed this rate to "Joule for Developers"; the published rate applies to Joule for Consultants. [S01]

CLIENT-SPECIFIC VALIDATION REQUIRED The monetary price of an AI Unit is not published on SAP's pricing page; it is quoted. Secondary sources citing SAP Learning material describe tiered per-step consumption for different agent classes, which differs from the flat per-action rate shown on SAP's pricing page [S30]. Confirm in writing which metric, which rate and which allowance apply to your contract, and whether any AI allowance is bundled with your FUEs.

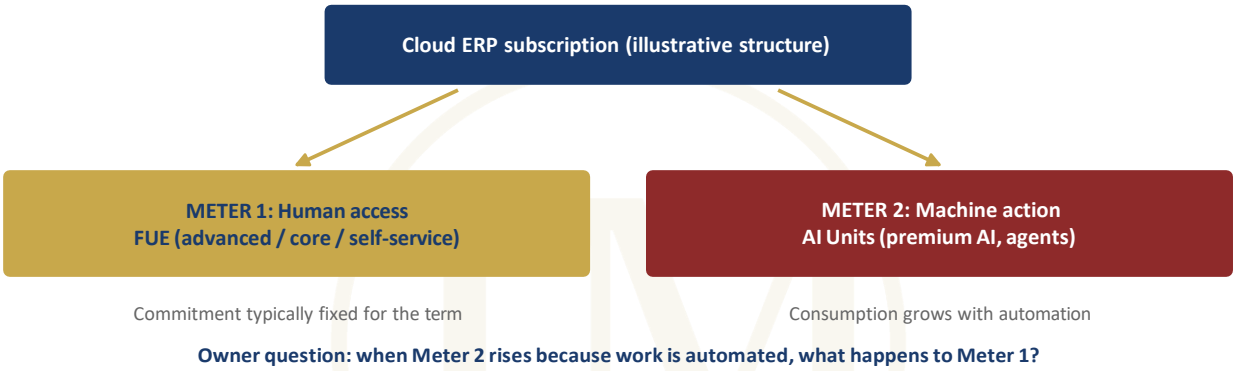


Figure. The two-meter structure. The owner question is how the meters interact over the contract term.

The two-meter problem

SOURCE-DOCUMENT POSITION The source document argued that under RISE the vendor "wins on both sides": FUE revenue stays flat as people are replaced, while AI Unit revenue grows as agents take over their work.

ADVISORY INTERPRETATION That outcome is possible, but it is not inevitable, and it is not necessarily unfair - the owner is buying new capability. The real risk is structural: if the FUE commitment is sized for today's workforce and the AI Unit budget is unconstrained, the owner pays for human capacity it no longer uses while paying again for the machine capacity that replaced it. The protection is contractual design, not ideology.

Commercial levers to test (each requires validation against your contract)

- **Classification evidence:** run your own user-classification analysis before SAP runs theirs; remove unnecessary advanced authorisations.
- **Renewal and reduction terms:** the right to reduce committed quantities at renewal, and the notice periods that apply.
- **Conversion or reallocation rights:** the source document proposed negotiating the right to convert FUE value into AI Units as automation grows. Whether SAP will agree is a negotiation outcome, not a published entitlement.
- **AI Unit rate and overage:** fix the unit price for the term, cap overage and require monthly consumption reporting.
- **Expiry:** AI Units expire after 12 months [S01]; size purchases on measured consumption, not roadmap ambition.

- **Price escalation:** renewal uplift caps for both meters.
- **Scope of Base AI:** document which AI functions you are using that are included, so that they are not later re-sold as premium.

JM ETA ADVISORY PRINCIPLE | Never confuse technical capability with licence entitlement

A function may exist in your system, appear in a demo or be switched on in a sandbox and still require a licence, an AI Unit balance or a specific plan to use in production. Every "it is already there" statement needs a licence answer as well as a technical one.

PART II

The Architecture Decisions

RISE, deployment models, the upgrade clock, custom code, BTP and Ariba - the choices that lock in cost for seven years or more.

CHAPTER 5

RISE with SAP - What Are You Actually Buying?

RISE is neither good nor bad. It is a specific bundle of subscription, infrastructure and technical operations. The owner's job is to know exactly which parts of the problem it solves and which parts remain on the owner's desk.

What RISE is

VERIFIED FACT RISE with SAP is a subscription offering built around SAP cloud ERP. For the Private Edition, AWS documentation describes a single-tenant environment in which SAP owns and manages the hyperscaler account, holds responsibility for the overall service level, cloud operations and technical support, and is responsible for security of the cloud environment. The offering also includes business process intelligence, platform and analytics components and business network elements. [S19]

VERIFIED FACT In 2025 SAP began marketing S/4HANA Cloud Public Edition as "SAP Cloud ERP" and Private Edition as "SAP Cloud ERP Private"; the licensed product names in contracts were reported as unchanged. [S27]

CLIENT-SPECIFIC VALIDATION REQUIRED The exact bundle - which BTP entitlements, which process-intelligence tools, which network components, which service levels, which system sizes and which environments - varies by contract. List every bundled component in your order form and map each one to a named use case and owner. Bundled components that nobody uses are still paid for.

What RISE is not

The most expensive misunderstanding in RISE programmes is the assumption that "SAP is now responsible" for things that remain the owner's responsibility. SAP's own Roles and Responsibilities documents are explicit on several points:

VERIFIED FACT For technical issues SAP provides support, but application-related problems must be resolved by the customer. Many application tasks are listed as Cloud Application Services available at additional charge, or to be performed by the customer. For system conversion, SAP executes the technical tool steps only it can perform, but overall responsibility for the conversion - functional conversion, pre- and post-conversion activities - remains with the customer. Customers do not receive operating-system access to managed servers. [S18]

- **RISE is not an implementation.** Design, configuration, data migration, testing, training and cutover are separate work, usually performed by an SI under a separate contract.
- **RISE is not application management.** Functional support, custom code support, interface monitoring and user support remain with the owner or are bought separately.
- **RISE is not a data clean-up.** Material masters, equipment records, functional locations and vendor records arrive in the cloud exactly as clean or as dirty as they left.
- **RISE is not automatic clean core.** Private Edition can carry classic custom code; whether it is clean depends on governance, not on hosting.
- **RISE is not a guarantee of functional fit.** Industry depth depends on edition, release and scope, not on the commercial wrapper.

Responsibility map - who does what

Area	SAP under RISE (per R&R documents)	Owner / SI	Owner-side watch point
Infrastructure and hosting	Provides and operates on the chosen hyperscaler [S19].	Selects region and data rules.	Confirm region, residency and sub-processors in writing.
OS, database and technical operations	Technical operations per R&R and SLA [S18].	No OS access to managed servers [S18].	Know which tasks are chargeable change requests.
Technical upgrade execution	Executes tool-based steps only SAP can perform [S18].	Owns functional conversion, testing and remediation [S18].	Budget testing and remediation at every upgrade.
Application incidents	Technical issues only [S18].	Resolves application problems or buys services [S18].	Price application support before signing.
Custom code	Hosts it.	Owns remediation, testing and retirement.	Maintain a custom code register with owners.
Master data	Not in scope.	Owns quality, governance and stewardship.	Fund data work separately and first.
Integrations	Provides platform services as contracted.	Designs, monitors and supports interfaces.	Name an owner for every interface.
Security	Security of the cloud environment [S19].	Roles, authorisations, SoD, user access.	Access governance remains an owner control.
Change management and adoption	Not in scope.	Owner and SI.	Usually under-budgeted in heavy industry.

The implications, one by one

Deployment

For large heavy-asset enterprises, RISE conversations typically centre on Private Edition, because it retains the full functional scope and extensibility associated with S/4HANA. The deployment decision therefore is less about "cloud or not" and more about who operates the infrastructure, where it runs and how much control the owner surrenders over timing and access.

Subscription

RISE converts licence ownership and annual maintenance into a recurring subscription for a defined term. That can improve predictability and reduce infrastructure capital expenditure. It also means the owner stops accumulating a perpetual right to use software and starts renting it, which changes the negotiating position at every renewal. Model the renewal as carefully as the first term.

Infrastructure and application responsibility

SAP takes infrastructure and technical operations; the owner keeps the application. In practice, that means the owner's IT function shifts from running servers to governing a vendor, while the functional support organisation - often the largest cost - is unchanged unless separately contracted.

Upgrades

VERIFIED FACT Customers on S/4HANA Cloud Private Edition are not forced to upgrade while their release remains in mainstream maintenance; Public Edition is upgraded by SAP twice a year and those upgrades are mandatory. [S14]

The upgrade decision is therefore still the owner's in Private Edition - but the cost of each upgrade (testing, remediation, business validation) is also still the owner's, and the maintenance clock described in Chapter 6 still applies.

Customisation and clean core

Private Edition permits classic ABAP modifications and custom objects. SAP nevertheless expects RISE customers to move towards clean core over time, and the cost of every upgrade will rise with the volume of unmanaged custom code. RISE does not remove the custom code question; it prices it into every future upgrade.

BTP

RISE bundles typically include some platform entitlement. BTP services are consumed against credits or subscriptions and create operational dependencies of their own (Chapter 9). The owner should know what is included, what consumption has been forecast and what happens when the included amount is exhausted.

AI

VERIFIED FACT Joule Base is included at no additional cost in SAP cloud subscriptions, and Joule has been available in S/4HANA Cloud Private Edition from release 2021, with new RISE contracts automatically entitled according to SAP community guidance. [S01, S08]

Premium AI and agents are consumed through AI Units (Chapter 4). RISE therefore gives access to SAP's AI roadmap, but not an unlimited right to use it.

Licensing

User access is licensed in FUE (Chapter 4). The sizing, classification and renewal terms of the FUE commitment are the single largest commercial exposure in most RISE contracts, and they must be modelled against the automation plan.

Commercial lock-in and exit

Every multi-year subscription creates lock-in; the question is whether it is priced and bounded. Exit considerations that should be answered before signature include: in what format and timeframe data is returned; what transition assistance is available and at what cost; what rights remain to any previously owned perpetual licences that were converted or credited; how integrations and BTP extensions would be re-hosted; and what the notice periods are.

CLIENT-SPECIFIC VALIDATION REQUIRED Conversion credits for existing licence investment, transition terms and exit assistance are contract-specific. Figures quoted in market commentary for historic conversion credits could not be verified for this guide and should not be relied upon.

Five-year and seven-year TCO - the structure that must be filled in

This guide does not publish costs; no reliable public price list exists for most of these lines. It does insist that every line below is populated with your own numbers for both RISE and the alternatives before a decision is taken.

Cost line	RISE (Private)	On-Premise / owner hosting	Notes
Software subscription or licence + maintenance	Subscription (FUE-based)	Licence (owned) + annual maintenance	Include renewal uplifts.
Infrastructure and hosting	Included in subscription scope	Owner capex/opex or hosting contract	Include DR and non-production.
Technical operations (Basis, DB, OS)	Largely included; some chargeable	Owner team or managed service	Map against R&R document.
Application management	Owner or purchased	Owner or purchased	Often the largest line.
Upgrades (testing, remediation)	Owner cost each cycle	Owner cost each cycle	Scales with custom code volume.
BTP and integration services	Included amount + consumption	Subscription or consumption if used	Forecast consumption.

Cost line	RISE (Private)	On-Premise / owner hosting	Notes
AI (AI Units, premium packages)	Consumption	Depends on eligibility and plan	Cap and forecast.
Data migration and clean-up	Owner / SI	Owner / SI	Required either way.
Security, compliance, audit	Shared	Owner	Include audit evidence effort.
Exit and transition	Contract-dependent	Contract-dependent	Price it now, not later.

The RISE decision framework

RISE is more likely to fit when the owner wants to exit infrastructure operations, accepts SAP-operated hosting under its data and regulatory rules, has a credible plan to reduce and govern custom code, values access to SAP's cloud-delivered roadmap and has modelled FUE and AI consumption against its automation plan.

RISE is less likely to fit when regulatory or national-infrastructure rules require owner control of the environment that SAP-operated hosting cannot satisfy, when the owner already runs infrastructure efficiently, when upgrade timing must be tied tightly to turnaround and shutdown calendars that the owner cannot reconcile with the maintenance clock, or when the commercial terms cannot be made flexible enough for the automation plan.

In all cases, the decision should be taken on a filled-in TCO structure, a signed responsibility map, a data-readiness assessment and an exit plan - not on a roadmap slide.

EVIDENCE REQUIRED | Before signing RISE

- Order form with every bundled component mapped to a use case and owner.
- Roles and Responsibilities document version referenced in the contract, with chargeable tasks identified.
- FUE sizing built from your own user-classification analysis.
- AI Unit rate, allowance, overage and expiry terms in writing.
- Upgrade cost model for two cycles, including custom code remediation and testing.
- Data residency, sub-processor and access terms validated by your legal and security teams.
- Exit terms: data return format, timelines, assistance and cost.

CHAPTER 6

Public vs Private vs On-Premise

There is no universal winner. There is only the right fit for a specific scope, a specific set of control requirements and a specific total cost.

VERIFIED FACT S/4HANA Cloud Public Edition is delivered as a multi-tenant service in which customers share the infrastructure and SAP upgrades the system twice a year; Private Edition provides a dedicated single-tenant environment; On-Premise is installed and operated by or for the owner. [S14, S28]

SOURCE-DOCUMENT POSITION The source working document argued that Public Edition is "useless for the Energy industry" and that Private Edition is effectively the old customised system wrapped in a subscription.

ADVISORY INTERPRETATION Both statements are too absolute. Public Edition is designed for standardised processes and a fixed extensibility model; it can be a sensible choice for subsidiaries, trading entities, shared-service or corporate functions and two-tier ERP landscapes. It is often a poor fit for the operating core of a complex asset-intensive business, where industry-specific depth and extensibility are needed. Private Edition is a genuine cloud operating model with SAP-operated infrastructure, but it does leave the application and its custom code with the owner. A fit-gap by process, not a generalisation, should decide.

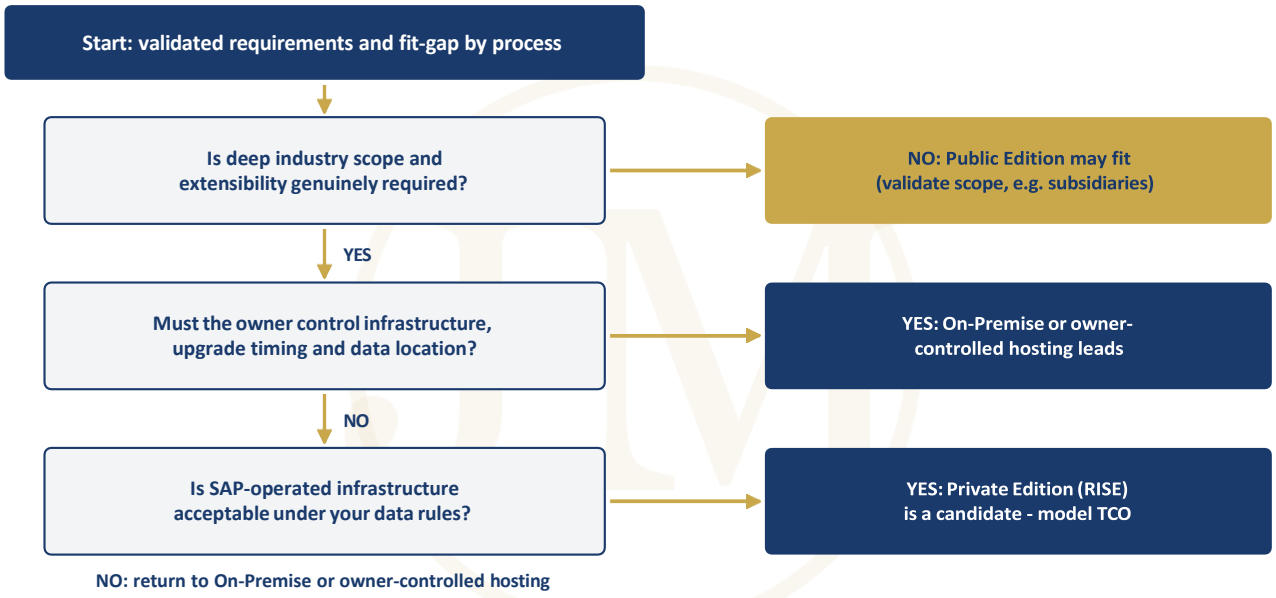
Dimension	Public Edition	Private Edition	On-Premise
Customisation	Released extensibility only; no classic modifications.	Classic ABAP and extensions permitted; clean core expected over time.	Full owner control; clean core governance still advisable.
Upgrade control	SAP upgrades twice yearly; mandatory [S14].	Owner chooses timing within mainstream maintenance [S14].	Owner chooses timing within maintenance terms.
Infrastructure control	SAP-operated, shared.	SAP-operated, dedicated [S19].	Owner or owner's provider.
Data sovereignty	Depends on available regions and terms.	Depends on region, hyperscaler and terms.	Highest owner control; owner carries the security burden.
AI / Joule access	Joule included with cloud subscription; premium via AI Units [S01].	Joule available from 2021 release; setup required [S08].	Limited; eligibility tied to RISE commitments and plans [S05].
BTP dependency	High for extensions.	Moderate to high.	Optional; used where valuable.
Legacy compatibility	Low; re-implementation typical.	High; conversion possible.	High.
Complex energy processes	Validate scope carefully.	Full S/4HANA scope incl. industry solutions (validate).	Full scope (validate licences).
EAM complexity	Validate depth for heavy maintenance.	Full EAM scope.	Full EAM scope.
Integration	Released APIs; cloud integration.	Released and classic interfaces.	Full range; policy still applies [S09].
Operational control	Lowest.	Shared.	Highest.
Commercial model	Subscription (FUE).	Subscription (FUE).	Licence + maintenance.
Five-year TCO	Model: subscription + adoption.	Model: subscription + app management + upgrades.	Model: licence/maintenance + infra + ops + upgrades.
Seven-year TCO	Include renewal terms.	Include renewal and one or two upgrades.	Include hardware refresh and upgrades.
Exit considerations	Data return; re-implementation risk.	Data return; re-hosting of extensions.	Licence rights retained; support options.

When each may be appropriate

Public Edition may be appropriate where processes are genuinely standard, the entity is small or mid-sized relative to the group, the owner wants SAP to carry the full operational load and the organisation is prepared to adopt SAP best practice rather than adapt the system. Examples in the energy sector might include a marketing subsidiary, a new-energy venture or a corporate services entity - subject to a fit-gap.

Private Edition may be appropriate where the owner needs the full S/4HANA and industry scope, is willing to have SAP operate infrastructure, can satisfy data and regulatory requirements in the available regions, and has a credible programme to govern custom code and upgrades.

On-Premise (including owner-controlled hosting) may be appropriate where regulatory, security or national-infrastructure requirements demand owner control, where the owner has efficient infrastructure operations, where upgrade timing must align tightly with operational calendars, or where the commercial case for subscription does not hold over seven years. The owner must then plan for maintenance terms, security patching and a realistic view of which SAP AI capabilities will be available.



Every branch ends in the same place: five- and seven-year TCO, exit terms and a data-readiness test.

Figure. A simplified deployment decision tree. Real decisions are made by process and by entity, and many groups run more than one model.

CHAPTER 7

The Upgrade Clock

Maintenance dates are used to create urgency. The owner should know exactly what the dates mean, what they do not mean and who actually does the work.

The facts

VERIFIED FACT Beginning with S/4HANA 2023, SAP moved to a two-year release cycle with seven years of mainstream maintenance per release, up from five years, with feature packs planned every six months in the first two years of a release. SAP also stated that at least one S/4HANA release will be in maintenance until 2040. [S12, S13]

VERIFIED FACT When mainstream maintenance ends, the release moves into customer-specific maintenance for customers with a valid support contract; support does not simply stop. [S15]

VERIFIED FACT Mainstream maintenance for SAP ECC 6.0 enhancement packages 6 to 8 ends at the end of 2027. Optional extended maintenance is available to the end of 2030 at an additional two percentage points on the maintenance fee. SAP has offered an "SAP ERP, private edition, transition option" for 2031 to 2033 for selected large and complex customers moving through RISE; SAP's chief executive stated this is not a prolongation of the 2027 and 2030 deadlines. [S16, S17]

SOURCE-DOCUMENT POSITION The source document described the upgrade as annual for Private Edition and quoted penalties of "2% to 4%" for extended maintenance. The release cadence is now two-yearly [S12], and the published extended maintenance premium for ECC is two percentage points [S16, S17]; any other figure must be validated against your contract.

What SAP cannot do - and what it can

SAP cannot physically push an upgrade into an owner-operated on-premise system. It can, however, change the support you receive once a release leaves mainstream maintenance - fewer or no new fixes, legal changes and security corrections for that release. For regulated energy businesses, operating without current legal and security corrections is rarely acceptable for long. The maintenance clock therefore is a real constraint, but it is a planning constraint, not a cliff edge.

Who actually does the upgrade work

Boards often assume that "a technical upgrade" is SAP's responsibility. It is not. SAP provides the software and the tools. In an on-premise estate, the owner and its SI or managed-service provider run the upgrade. Under RISE, SAP performs the tool-based technical steps that only it can execute, while the overall responsibility for conversion and the functional work remains with the customer [S18].

Workstream	Typically performed by	Owner-side control
Technical execution (upgrade tools, database, kernel)	SI / MSP on-premise; SAP for tool steps under RISE	Approve runbook, downtime window and rollback plan.
Custom code adjustment (modification adjustment and remediation)	Owner developers or SI	Custom code register; retire unused objects first.
Integration regression	Owner / SI	Interface inventory with named owners.

Workstream	Typically performed by	Owner-side control
Business validation (UAT)	Owner business users	Scripts for critical EAM, procurement and finance flows.
Data validation	Owner data stewards	Reconciliation gates before production cutover.
Security and authorisations	Owner / SI	Re-test SoD and critical access.

Third-party support and the statistics you may be shown

SOURCE-DOCUMENT POSITION The source document cited statistics attributed to third-party support providers and consultancies - for example, that over 60% of early S/4HANA adopters did not upgrade, that 15% to 20% of heavy-asset enterprises freeze their ERP and move to third-party support, and that 80% of compliant customers choose purely technical upgrades.

CLIENT-SPECIFIC VALIDATION REQUIRED These figures could not be traced to primary, methodologically transparent sources for this guide and are not reproduced as facts. Third-party support is a legitimate market option with real trade-offs - it typically cannot deliver new SAP code corrections, legal updates or security patches from SAP - and should be evaluated on your own risk profile, contract terms and regulatory obligations, not on survey headlines from any party with a commercial interest.

JM ETA ADVISORY PRINCIPLE | Do not let a support date dictate an architecture

A maintenance deadline is a reason to plan, not a reason to sign. The options - upgrade within the current model, pay for extended maintenance where available, move to a different deployment model, or change support provider - each have a cost and a risk. Model them side by side, with your turnaround calendar, before any vendor turns a date into a deadline.

CHAPTER 8

The Z-Table Question

Are Z tables and ABAP automatically technical debt? No. But uncontrolled customisation is - and clean technology cannot compensate for dirty business data.

Few phrases have moved more money in SAP programmes than "technical debt". It is used to justify re-implementation, cloud migration, BTP rebuilds and large SI work packages. It is sometimes accurate. It is sometimes a label applied to legitimate business logic because rebuilding it is profitable. The owner needs a more precise test.

What SAP itself now says

VERIFIED FACT In 2025 SAP evolved its three-tier extensibility model into a "clean core level" concept that grades extensions from A to D. Level A covers extensions using only released, stable interfaces - either on-stack in S/4HANA using the ABAP Cloud development model or side-by-side on SAP BTP. Level B, which uses documented classic APIs, is also considered clean core. Level C is conditionally clean and Level D - relying on objects SAP does not recommend - is not clean core. [S20]

ADVISORY INTERPRETATION Two consequences follow. First, SAP's own framework does not equate "custom" with "debt": custom logic built inside the core with released technology can be graded at the cleanest level. Second, location is not the test - a badly designed BTP application calling unstable interfaces is not cleaner than a well-built on-stack extension. The test is stability of the interfaces used, governance of the code and justification of the requirement.

Why uncontrolled customisation does create debt

Custom code becomes debt when nobody can say why it exists, who owns it, whether it is used, what it touches and how it will behave at the next upgrade. Heavy-asset estates frequently carry thousands of custom objects, a significant share of which are unused, duplicated or written to work around poor master data rather than to deliver differentiation. Every one of those objects is tested, adjusted and supported at each upgrade.

Why some customisation is legitimate differentiation

Asset-intensive industries have genuinely specialised requirements: joint-venture and production-sharing arrangements, hydrocarbon allocation, integrity and inspection regimes, permit-to-work and isolation links to maintenance, competency checks before work release, specialised traceability for pressure-retaining and safety-critical components, and local content obligations. Some of these are covered by SAP industry solutions and standard functionality; some are not. Where they are not, custom logic may be the correct, least-cost answer.

CLIENT-SPECIFIC VALIDATION REQUIRED Before accepting that a heavy-industry requirement needs custom code, confirm whether the relevant SAP industry solution, standard function or later feature pack covers it in your release and whether you are already licensed for it.

Governance, not ideology

- **Custom code register:** every object with an owner, business justification, usage evidence and clean core level.
- **Retirement first:** unused objects are removed before any migration or upgrade is estimated; paying an SI to migrate dead code is pure waste.
- **Quality gates:** automated code checks configured for clean core levels, with exemptions approved by an architecture authority, not by the developer.
- **Design authority:** every new custom request passes through the governance sequence in Chapter 1.
- **AI-assisted maintenance:** AI coding assistants can help explain and document legacy code; the source document's claim of 60% to 70% cost reduction is unverified - measure it in your own pilot.

Four myths to retire

"Moving it to BTP makes it clean." Moving bad logic to another platform relocates the problem and adds integration, latency and a second lifecycle. **"Cloud is better."** Cloud is a delivery and operating model; it is better only when it serves the requirement at a lower lifetime cost or risk. **"Standard means fit."** Standard means SAP's model of the process; fit must be proved against yours. **"Technical debt is the main problem."** In most heavy-asset estates the larger debt is in the data - duplicated materials, broken hierarchies, missing MPNs - which no code strategy can fix.

JM ETA ADVISORY PRINCIPLE | Customisation is not automatically the enemy

Customisation is not automatically the enemy. Uncontrolled customisation is.

And clean technology cannot compensate for dirty business data.

CHAPTER 9

BTP and Clean Core

"Everything can now be moved to BTP" is a sentence that should never be accepted without architecture analysis.

What clean core is intended to achieve

Clean core aims to keep the ERP upgradeable by ensuring that extensions use stable, released interfaces rather than modifying SAP code or depending on internal objects that may change. For owners, the benefit is real: lower upgrade effort, faster adoption of innovation and less dependence on specialist knowledge of historic modifications.

SOURCE-DOCUMENT POSITION The source document argued that SAP promotes clean core primarily to secure predictable cloud revenue, control the AI layer and reduce its own support costs.

ADVISORY INTERPRETATION Commercial motives and genuine engineering benefits can coexist. The owner does not need to decide SAP's motives; it needs to decide whether a specific clean core action creates value for its own estate at an acceptable cost.

Where BTP is useful

- **Side-by-side applications** that serve users outside the ERP (contractors, suppliers, field crews) or combine SAP and non-SAP data.
- **Integration services** connecting SAP with cloud applications, networks and external parties.
- **Workflow and process automation** spanning several systems.
- **Analytics and data services**, where data is deliberately replicated for analysis.
- **AI services and agent tooling** within SAP's ecosystem.
- **Mobile and partner-facing experiences** that should not load the ERP directly.

What should normally remain in the core

Logic that must be transactionally consistent with postings, validations that must run before a document is saved, performance-critical processing close to large data volumes, and controls that must not be bypassable by an interface outage. The source document gave examples such as high-frequency allocation and pricing calculations; the underlying principle is sound - when logic and data must move together, separating them adds latency, failure points and reconciliation effort.

VERIFIED FACT SAP's own clean core guidance places on-stack extensions built with ABAP Cloud at Level A, the same level as side-by-side BTP extensions; clean core does not require moving logic off the ERP. [S20]

The hidden costs of side-by-side

Consideration	What to ask
Cost	Which BTP services, which metrics, what forecast consumption, what happens after included entitlements are used?
Complexity	How many new components, runtimes and skills does this add? Who supports them?
Operational dependency	If BTP or connectivity fails, which business process stops? Is that acceptable during a shutdown?
Performance	How many API calls per transaction? What latency is acceptable? Has volume testing been done?
Data movement	What data leaves the ERP, where is it stored and for how long? Is it duplicated?
Security	Which identities, which connectivity, which audit logs, who reviews them?
Lifecycle	Who updates runtimes and libraries? What is the test cycle when BTP services change?
Governance	Who approves new BTP applications, and against which architecture principles?

RED FLAG | "Everything can now be moved to BTP."

Technically, much can be built on BTP. Whether it should be depends on performance, consistency, cost, dependency and support.

What to ask next: for each object proposed for BTP, show the clean core level today, the level after the move, the integration pattern, the consumption forecast, the support owner and the reason on-stack released extensibility was rejected.

CHAPTER 10

SAP Ariba and On-Premise S/4HANA

Can SAP Ariba coexist with On-Premise S/4HANA or ECC? Yes. A cloud procurement front-end does not require a cloud ERP.

The hybrid architecture

SAP Ariba solutions are delivered as cloud services. The ERP that holds the material master, supplier master, financial ledger and asset hierarchy can remain on-premise or in owner-controlled hosting. The two are connected through SAP-provided integration components.

VERIFIED FACT SAP documents the integration of SAP ERP and SAP S/4HANA with SAP Ariba applications and SAP Business Network through "SAP Integration Suite, managed gateway for spend management and SAP Business Network" - the renamed SAP Ariba Cloud Integration Gateway - including add-on installation for SAP ERP and S/4HANA systems. SAP states its documentation applies to both SAP ERP and SAP S/4HANA. [S21]

VERIFIED FACT SAP Cloud Connector runs as an on-premise agent and acts as a reverse-invoke proxy between the on-premise network and SAP cloud services, so that inbound firewall ports do not need to be opened; it provides fine-grained access control and audit logging of inbound traffic and configuration changes. [S22]

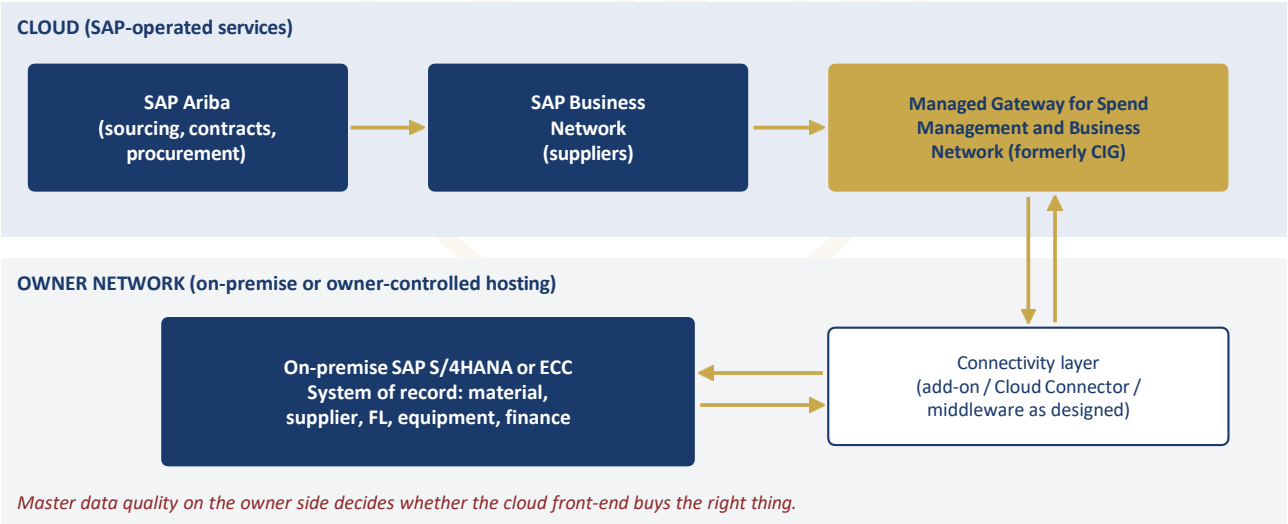


Figure. Hybrid procurement architecture. The exact connectivity pattern depends on scope and design and must be validated for your landscape.

SOURCE-DOCUMENT POSITION The source document described this hybrid as one of the most common procurement architectures in heavy-asset and EPC companies. That is consistent with practitioner experience but is not a verified statistic.

What flows where

Object	Typically mastered in	Direction	Owner-side risk
Material master and catalogue content	ERP (and catalogue management)	ERP → Ariba	Duplicates and missing MPNs push users to free text.
Supplier master	ERP / supplier management / MDG (design choice)	Bidirectional by design	Two masters with different truths; weak onboarding controls.
Purchase requisitions	ERP or Ariba (scenario-dependent)	Either	Account assignment to wrong equipment or cost object.
Purchase orders	Ariba or ERP (scenario-dependent)	Either	Approval limits configured differently in two systems.
Order confirmations, ship notices	Business Network	Network → ERP	Message failures unnoticed during critical deliveries.
Goods receipt and QA/QC	ERP	ERP → Ariba/Network	Certificates not linked to the received material.
Invoices	Network / Ariba → ERP	Network → ERP	Tolerance and three-way match rules not aligned.
Contracts and pricing	Ariba (contracts)	Ariba → ERP	Contract prices not reaching the transaction.

Security, ownership and support

A hybrid architecture has more than one support boundary. SAP operates the cloud services; the owner operates the ERP and the connectivity components on its side; the SI may have built the integration. When a purchase order for a critical spare fails to reach a supplier, somebody must see it, own it and fix it within the operational timeframe. The owner should insist on an interface inventory, monitoring responsibilities, alerting thresholds and escalation paths before go-live.

Licensing and commercial implications

Ariba subscriptions are separate from ERP licensing, and integration components and platform services may carry their own terms. Running Ariba with on-premise ERP does not require a RISE contract. Where a vendor or SI suggests otherwise, ask for the specific product requirement in SAP documentation.

CLIENT-SPECIFIC VALIDATION REQUIRED Confirm which Ariba modules, network fees, integration components and platform services are included in your subscription, and whether any are metered by document volume or spend.

The heavy-asset trap

Ariba is a window onto the supplier market. It is not a foundation. If the material master behind it is full of duplicates and vague descriptions, a technician looking for a specific valve or gasket will not find it in the catalogue and will raise a free-text request. Free text bypasses contracts, weakens price validation and breaks the traceability chain from purchase to installation. The result is a modern cloud procurement interface producing the same leakage as the paper process it replaced - only faster.

> **JM ETA ADVISORY PRINCIPLE | Ariba is a window, not a foundation**
>
> Connecting Ariba to an on-premise ERP is technically straightforward. Making it buy the right safety-critical part, from the right supplier, at the contracted price, with the right certificate, depends entirely on the owner-side master data. Fix the catalogue before you open the window.

PART III

The Foundation Decides Everything

Asset data, master data, procurement governance and data ownership - the owner-side work that no vendor, SI or AI product can do for you.

CHAPTER 11

EAM in Asset-Intensive Industry

AI cannot rescue a broken asset foundation. Automating bad data does not create good decisions. It creates bad decisions faster.

This chapter is written for Oil and Gas, refining, petrochemicals, power generation, utilities, mining, LNG, offshore, terminals, heavy manufacturing and EPC. In these industries, Enterprise Asset Management is not a module; it is the operating memory of the plant. It records what is installed, where, what it is made of, how it fails, how it is maintained, what spares it needs and who is allowed to work on it. Every AI promise in maintenance and supply chain - predictive maintenance, autonomous planning, intelligent procurement, digital twins - depends on that memory being accurate.

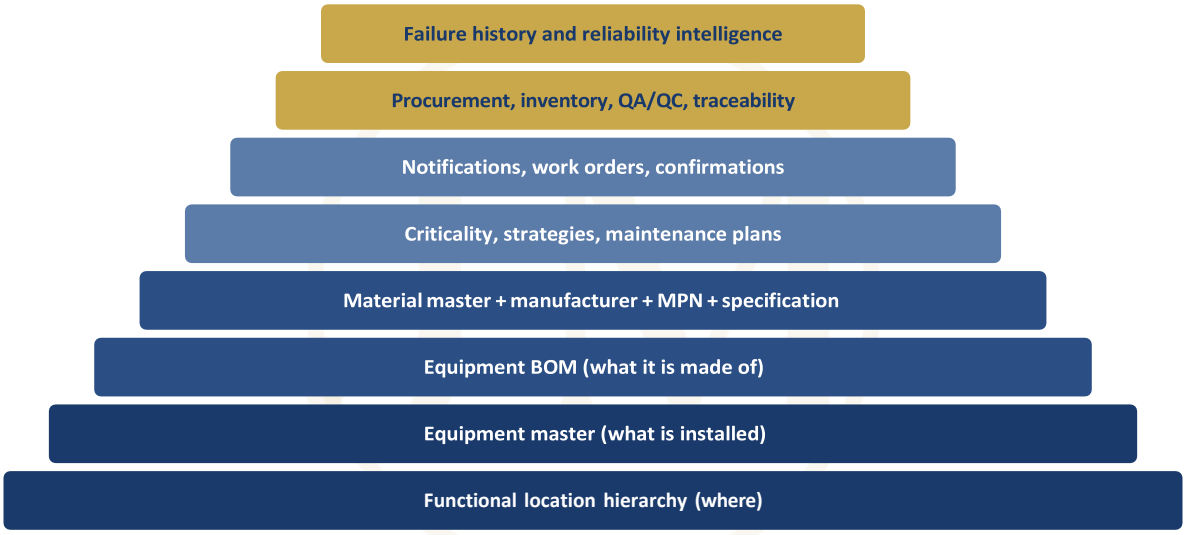


Figure. The asset data foundation. Each layer depends on the one below it; an error at the base propagates upwards into every plan, order and purchase.

The foundation, layer by layer

Functional locations and the asset hierarchy

The functional location hierarchy describes where things are and what function they serve: site, unit, system, sub-system, position. It must reflect the physical plant and the way operations, maintenance and engineering think about it. Common failures include hierarchies copied from project structures that no longer match the operating plant, inconsistent levels between sites, and positions without the technical object types that drive maintenance and reporting. When the hierarchy is wrong, costs roll up to the wrong place, criticality is assigned to the wrong object and reliability analysis compares unlike with unlike.

Equipment master

Equipment records describe the individual items installed in those positions - with manufacturer, model, serial number, specifications, classification and installation history. When a pump is swapped during a turnaround, the equipment moves and the history follows it. Poorly maintained equipment masters lose that history, which destroys the failure record that reliability engineering and AI both need.

Equipment BOM

The equipment bill of materials links each item to the spare parts that maintain it. A correct BOM is what allows a planner - or an agent - to reserve the right parts for a work order without guessing. Missing or generic BOMs are one of the most common causes of wrong parts, emergency purchases and extended outages in heavy industry.

Material master, manufacturer and manufacturer part number

The material master is where maintenance and procurement meet. For MRO materials, the identity of an item is defined by its specification, its manufacturer and its manufacturer part number (MPN). Duplicate materials created with different descriptions for the same MPN split stock, hide availability, break contract pricing and inflate inventory. A material record without a verified MPN cannot be matched reliably by a human, a catalogue or an AI agent.

Critical spares

Critical spares are held because their absence would stop production, compromise safety or breach a regulatory obligation. They should be linked to the equipment they protect, justified by criticality and lead time, and reviewed periodically. Where that link is missing, critical spares are either over-stocked "just in case" or silently consumed for non-critical work.

Maintenance strategies, plans, notifications and work orders

Strategies and maintenance plans convert engineering knowledge into scheduled work. Notifications capture what was found; work orders capture what was done, with what materials, by whom and at what cost. The value of this data for reliability and AI depends on structured completion: damage, cause and activity codes, measurement readings and genuine failure descriptions rather than "job completed".

Procurement, inventory and contractor maintenance

Maintenance demand becomes procurement demand through reservations, requisitions and service entries. Contractor maintenance - increasingly dominant in heavy industry - must still write back to the equipment history, or the owner loses the knowledge of its own plant to its contractors.

QA/QC and traceability

For safety-critical and pressure-retaining components, the owner must be able to trace an installed item back to its material test certificate and inspection records. That chain runs from the purchase order, through goods receipt and quality inspection, to the equipment and functional location where the item was installed. If that chain is not recorded in the system, procurement controls protect the purchase order - not the plant.

Asset criticality, failure history and reliability data

Criticality ranking determines where effort and spares investment go. Failure history and reliability data - mean time between failures, failure modes, repair times - are what predictive and prescriptive AI actually learn from. They are only as good as the structured data captured on every notification and order over many years.

The failure chain when the foundation is weak

Data object	What good looks like	Typical failure	Consequence when automated
Functional location	Mirrors the physical plant; consistent levels across sites.	Copied from project WBS; inconsistent.	Agent assigns work and cost to the wrong asset.
Equipment	Unique, classified, with manufacturer, model and serial.	Duplicates; generic records; missing history.	Failure patterns invisible; predictions unreliable.
Equipment BOM	Complete, validated against OEM data.	Missing or generic BOMs.	Agent reserves wrong parts or none.
Material master	One record per item; structured description; classification.	Duplicates; free-text descriptions.	Agent buys what is already in stock.
Manufacturer and MPN	Verified MPN linked to each material.	Missing or unverified MPN.	Agent cannot match supplier offers reliably.
Critical spares	Linked to protected equipment; justified.	No link; unreviewed stock.	Agent consumes or re-orders critical stock incorrectly.
Maintenance history	Structured damage, cause, activity codes.	"Job done" completions.	AI learns nothing useful - or learns the wrong thing.
Traceability	MTC linked at GR to material and installed location.	Paper certificates, unlinked.	Non-conforming items installed without detection.

JM ETA ADVISORY PRINCIPLE | AI cannot rescue a broken asset foundation

Automating bad data does not create good decisions. It creates bad decisions faster.

A safety or operational standard that your asset data cannot support is not a control. It is an intention with a document number.

CEO QUESTION | Three questions for the Plant Director and EAM Head

- What percentage of our safety-critical equipment has a complete, validated BOM?
- Can we trace a pressure-retaining component installed last year back to its material test certificate from the system record alone?
- How many duplicate materials exist for the same MPN, and what stock value sits in them?

If these answers are unknown, the next investment should be in the foundation, not in another layer on top of it.

CHAPTER 12

Before AI - Fix the Data

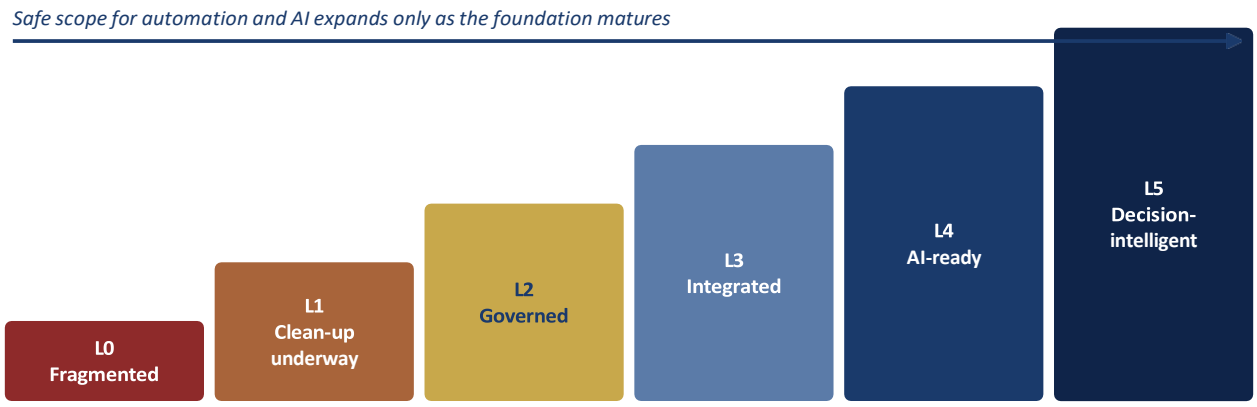
A governed data master is the first requirement for Ariba, for S/4HANA, for any ERP - and for any business, even one without an ERP.

SOURCE-DOCUMENT POSITION The source document reached a conclusion that this guide adopts without reservation: master data quality is a precondition for business continuity on any system, and would be even without an ERP. Software inherits the data; it does not clean it.

What AI actually depends on

Data domain	Why AI depends on it
Material master quality	Matching, reordering, substitution and price comparison all depend on one clean record per item.
Vendor master quality	Supplier selection, payment, risk screening and duplicate detection require unique, verified supplier records.
Equipment master quality	Failure prediction and work planning need correct identity and history per installed item.
Functional-location hierarchy	Aggregation, criticality and cost analysis depend on a hierarchy that reflects the plant.
Manufacturer and MPN data	The only reliable key for matching parts across catalogues, suppliers and sites.
Specifications and classification	Enable search, substitution and technical validation without human interpretation.
Equipment BOM	Links demand from assets to the materials required.
Maintenance history	The training and evidence base for reliability analysis and predictive models.
Procurement history	Price benchmarks, lead times and supplier performance.
Inventory records	Availability decisions and reorder logic; errors trigger unnecessary purchases.

The JM ETA data maturity model



Most heavy-asset owners must evidence at least Level 2 in the target domain before agentic automation.

Figure. Six levels of data maturity. The level should be assessed per domain and per site; averages hide risk.

Level	What it means	Evidence you would see	What automation can safely do
Level 0 - Fragmented / unreliable	No owners, no standards; duplicates and gaps widespread; spreadsheets outside the system hold the real knowledge.	Frequent emergency purchases; stock found after buying; planners maintain private lists.	Nothing autonomous. Use AI only to analyse and profile the data.
Level 1 - Clean-up underway	A cleansing programme exists; standards drafted; backlog shrinking but no sustaining governance.	Project team cleaning data; new errors still entering.	AI-assisted cleansing proposals with human approval.
Level 2 - Governed	Named owners and stewards; creation and change controlled by workflow; quality measured.	Data quality KPIs reported; creation requests approved against standards.	Recommendations to humans; low-risk automation in defined areas.
Level 3 - Integrated	Master data consistent across ERP, EAM, procurement and engineering systems; single source per object.	No reconciliation spreadsheets; interfaces monitored.	Automated transactions within narrow categories and limits.
Level 4 - AI-ready	Complete, classified, traceable data with history; controls designed for agents.	Agent identities, logs and limits in place; audit tested.	Agentic execution for governed categories with exception routing.
Level 5 - Decision-intelligent	Data, models and governance continuously improve decisions; feedback loops from outcomes.	Decisions measured against outcomes; models retrained on governed data.	Broad agentic operation with human oversight of policy and exceptions.

MDG is a tool; governance is an organisation

SAP Master Data Governance can enforce workflow, validation and approval for master data. It cannot decide who owns the equipment master, what the naming standard for a gate valve is, or who is accountable when a duplicate supplier is created. Buying MDG without a governance model - owners, stewards, standards, decision rights and KPIs - is a classic example of buying the BUY step before the PROVE step.

EVIDENCE REQUIRED | Before any AI or agent proposal is approved

- A data maturity assessment for the specific domain the AI will touch, by site.
- Named data owners and stewards for that domain.
- Duplicate, completeness and accuracy metrics measured on real data, not on a sample chosen by the vendor.
- A remediation plan and budget that precede, not follow, the AI deployment.

CHAPTER 13

AI Agents in Procurement

An AI agent can run the procurement sequence end to end. The question is not whether it can - it is who controls the guardrails.

The conceptual case

The source document used a practical example: inventory of transformer oil at a utility plant falls below its reorder point, and an agent is tasked with replenishing it. The example is useful because it is ordinary. It is exactly the kind of high-volume, rules-based transaction that agents will be asked to handle first.

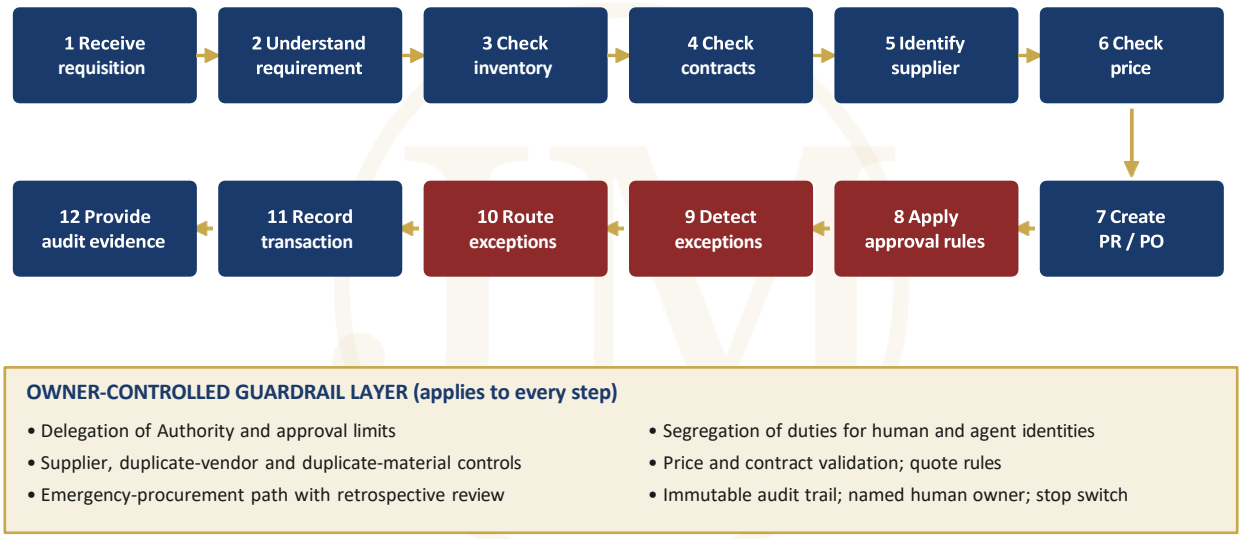


Figure. A twelve-step agent procurement flow. Steps 8 to 10 are where governance lives; the guardrail layer applies to every step.

1. **Receive a requisition** - from a reorder alert, a work order reservation or an unstructured request such as an email or PDF.
2. **Understand the requirement** - extract material, quantity, delivery location, account assignment and urgency; resolve it to a material master record.
3. **Check inventory** - across storage locations and sites, including stock in transit and reserved quantities.
4. **Check contracts** - identify framework agreements, scheduling agreements or catalogue items that already cover the need.
5. **Identify the supplier** - from the contract, source list or approved supplier list, not from the agent's own judgement.
6. **Check price** - against the contract condition and recent purchase history.
7. **Create the PR or PO** - with correct account assignment to equipment, order or cost object.
8. **Apply approval rules** - route through the Delegation of Authority exactly as a human-created document would be routed.

9. **Detect exceptions** - price variance, missing contract, new supplier, unusual quantity, split orders, urgency claims.
10. **Route exceptions** - to a named human role with the evidence attached; never resolve them silently.
11. **Record the transaction** - with the agent's identity, inputs and decision path stored.
12. **Provide audit evidence** - retrievable by internal audit without the agent's designer present.

ADVISORY INTERPRETATION In the source example, the agent encountered a price mismatch against the contract, opened the contract system, concluded the increase was authorised, adjusted the price and added a note. That is precisely the behaviour an owner should prohibit. An agent that resolves its own price exceptions has become an approver. The correct design is for the agent to gather the evidence and route the exception to the accountable buyer.

SOURCE-DOCUMENT POSITION The source document also presented a comparison table claiming agents complete purchase orders in 45 to 90 seconds with error rates below 0.5%, against 12 to 20 minutes and 5% to 8% for humans. These figures are not verified and are not used in this guide. Measure speed and error rates in your own controlled pilot.

Who controls the guardrails?

Control	What it must do when an agent is the actor
Delegation of Authority	Apply identically to agent-created documents; an agent never holds approval authority above the lowest routine threshold set by the board.
Approval limits	Hard limits per agent, per category, per supplier and per period - enforced in the ERP, not only in the agent's prompt.
Segregation of duties	The agent identity that creates cannot approve, receive or pay; agent identities are included in SoD analysis.
Integrity controls	Process design that removes individual discretion from high-risk decisions applies to agents as it does to people; monitor for repeated supplier patterns and threshold-splitting.
Supplier controls	Agents select only from approved suppliers; new supplier creation is always a human-governed process.
Duplicate vendors	Block transactions against suspected duplicates; route for steward review.
Duplicate materials	Detect near-duplicate materials before purchase; propose consolidation rather than buying.
Price validation	Tolerance rules against contract and history; variances are exceptions, not adjustments.
Contract validation	Contract validity, scope, value consumption and expiry checked before use.
Three-quote rules where applicable	Where policy requires competitive quotes, agents gather and document them; award remains with the designated human or committee.
Emergency procurement	A defined fast path with post-event review; "urgent" claims are logged and analysed for misuse.
Audit trail	Immutable logs of inputs, data consulted, rules applied and outputs, linked to each document.
Human approval	Mandatory above defined values, for new suppliers, for safety-critical materials and for any exception.
Exception management	Named owners, response times and escalation; unresolved exceptions stop the flow rather than defaulting to "proceed".

JM ETA ADVISORY PRINCIPLE | The DoA rule for the agent era

Delegation of Authority must be enforced identically inside the ERP and outside it - in workflow gates and system blocks, and in physical sign-off formats and committee minutes. A workaround in one channel invalidates the other. When an agent enters the process, it inherits the same rule: it is one more actor whose discretion the process must constrain.

CHAPTER 14

AI Does Not Replace Governance

The faster the machine executes, the more important the governance becomes.

Automation without governance is accelerated risk.

An AI agent executing an incorrect procurement decision at one transaction per minute is a problem. An AI agent executing a thousand incorrect transactions automatically is an enterprise event. The difference is not the intelligence of the agent; it is the speed and scale at which an unchecked error propagates before a human notices.

Traditional ERP controls were designed around human speed. A buyer might raise a few dozen orders a day; an approver reviews them; audit samples them after the fact. Agents remove the natural brake of human speed. Controls that relied on "someone would notice" no longer work.

The minimum governance architecture for agents

- **Named human owner** for every agent, accountable for its outcomes as a line manager is for a team.
- **Distinct, non-human identities** for agents, with least-privilege authorisations and inclusion in access reviews.
- **Transaction and value limits** enforced by the system of record, not by instructions to the agent.
- **Stop switch** that can suspend an agent immediately, with a tested fallback process.
- **Logging and explainability** sufficient for an auditor to reconstruct why each action was taken.
- **Monitoring** of volumes, exception rates, supplier concentration and value trends, with alerts.
- **Staged release** from recommendation, to execution with approval, to autonomous execution within limits - each stage evidenced before the next.
- **Periodic review** of agent performance, drift and control effectiveness by an independent function.

ADVISORY INTERPRETATION This is where the HSE principle familiar to every heavy-asset operator applies directly. The central function provides the standards, the tools and the reporting; each department head owns the outcome. For AI, IT and the vendor provide the platform; the process owner owns what the agent does in their process.

> **CEO QUESTION | Before any agent goes live**
>
> "Show me the agent's approval limits, who owns it, how we stop it, and the audit report our internal auditors will receive at the end of the first month."
>
> If any of the four cannot be shown, the agent is not ready for production.

CHAPTER 15

The Sovereign Data Moat

In an era when every competitor can buy the same software and the same AI models, the durable advantage is the proprietary data you own, govern and control.

SOURCE-DOCUMENT POSITION The source document borrowed the investment concept of an economic "moat" - a durable competitive advantage - and argued that for AI-era enterprises the moat is no longer the software bought but the data, governance and decision ownership retained. It cited the move towards sovereign AI in the UAE as an example.

ADVISORY INTERPRETATION For heavy-asset companies the argument is compelling on its own terms, without any geopolitical claim. Decades of operating, maintenance, failure, engineering and commercial data are unique to each owner. AI models are increasingly available to everyone; the data that makes them useful for a specific plant is not.

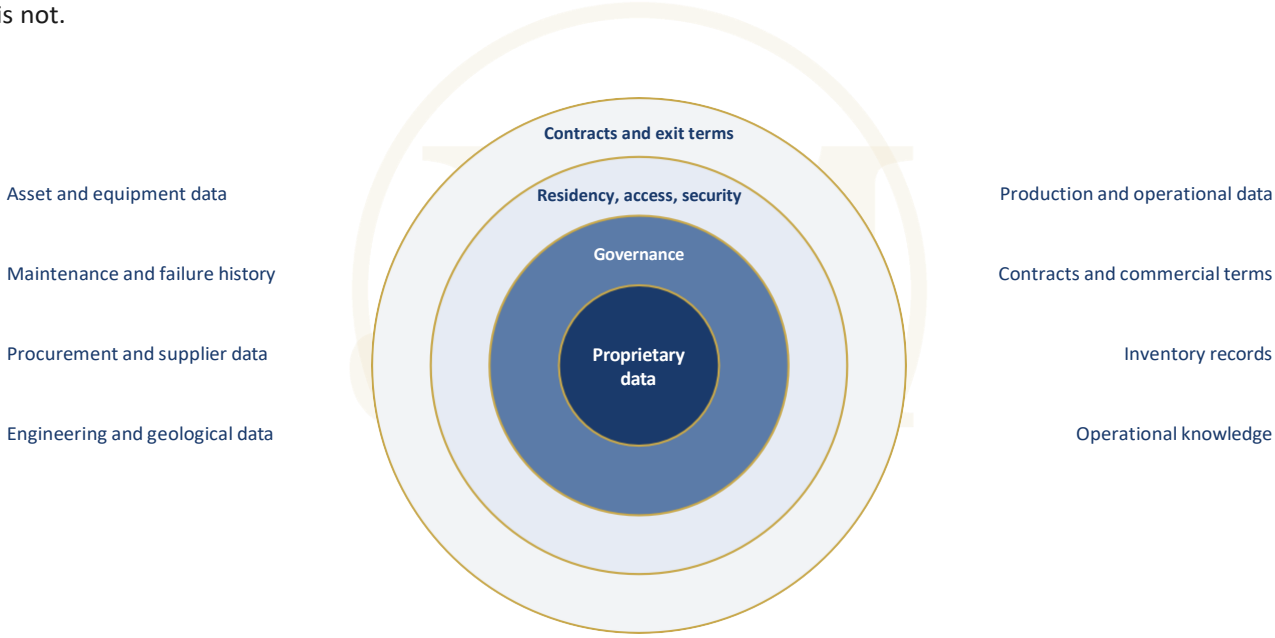


Figure. The sovereign data moat: proprietary data at the centre, protected by governance, by residency and security controls, and by contract terms on model use, portability and exit.

What is in the moat

Proprietary enterprise data includes operational and production data, asset and equipment records, maintenance and failure history, procurement and supplier performance data, engineering and design data, geological and reservoir data where relevant, contracts and commercial terms, and the operational knowledge embedded in procedures, job plans and lessons learned.

What the owner should decide deliberately

Consideration	Question to answer	Where the answer lives
Ownership	Who owns data we create, and data derived from it by vendor services?	Master agreement, data processing terms.
Governance	Who inside our organisation decides how each data domain may be used?	Data governance charter.
Residency	Where is data stored and processed, including backups and support access?	Service descriptions, region selection, sub-processor lists.
Access	Which vendor personnel and systems can access our data, and how is it logged?	Security terms, audit reports.
Retention	How long is data kept, and how is it deleted at end of contract?	Contract and data processing terms.
Model usage	Can our data or prompts be used to train or improve vendor models?	AI terms and conditions.
Security	Which certifications, controls and incident obligations apply?	Security annexes, certifications.
Portability	In what format, how fast and at what cost can we extract our data?	Exit and transition terms; API policy [S09].
Vendor dependency	Which decisions would stop if the vendor relationship ended tomorrow?	Architecture review; exit plan.

CLIENT-SPECIFIC VALIDATION REQUIRED Data residency and national data rules differ by country and sector and change over time. This guide does not make legal or geopolitical claims; obtain legal advice for your jurisdiction and map every data category against the contract terms above.

JM ETA ADVISORY PRINCIPLE | The tool is not the moat

The advantage is not the tool you buy. It is the data, the governance and the decision ownership you retain - and the contract terms that keep them yours.

PART IV

The Owner's Toolkit

Printable frameworks for the steering committee: the challenge framework, the CEO 100, the red-flag sentences, the value protection checklist and the golden rules.

CHAPTER 16

The SAP/SI Challenge Framework

For every proposed product, module, licence or customisation - twelve questions and the evidence each one requires. Print it. Take it into the room.

#	Gate	Question	Evidence that closes the gate
1	REQUIREMENT	What exact business problem are we solving?	Business requirement signed by the process owner, with volumes and measurable benefit.
2	STANDARD	What SAP standard capability was evaluated?	Fit-gap against our release and edition, with SAP documentation references.
3	CONFIGURATION	Why can configuration not solve it?	Configuration options tested and the reason each failed.
4	EXTENSION	Why can supported extensibility not solve it?	Assessment of key-user, developer and side-by-side options, with clean core level.
5	INTEGRATION	Why is integration required?	Data flow design, system of record per object, interface owner.
6	CUSTOM	Why is custom code required?	Justification, owner, clean core level, test and retirement plan.
7	PRODUCT	Why is another SAP product required?	Proof that owned products and standard scope cannot meet the need.
8	LICENCE	What exact licence is required?	Licence metric, quantity, term and the clause that grants the right.
9	COST	What is the five-year TCO?	Full cost model including integration, testing, support and consumption; seven-year view too.
10	RISK	What new dependency does this create?	Operational, technical, commercial and skills dependencies, with mitigations.
11	EXIT	How do we remove it later?	Exit steps, data return, notice periods and cost.
12	EVIDENCE	Show us the documentation.	Documents, not slides: SAP documentation, contract clauses, test results.

The framework is deliberately neutral. It does not presume that the proposer is wrong. It presumes that a decision of this size deserves documents. A proposal that passes all twelve gates should be approved with confidence; a proposal that cannot pass them should not be approved on confidence alone.

NOTE | How to use it in governance

- Attach the completed framework to every change request, product proposal and SI change order above an agreed threshold.
- Require the owner-side process lead, not the proposer, to confirm gates 1 to 3.
- Require the architecture authority to confirm gates 4 to 7 and 10.
- Require finance and procurement to confirm gates 8, 9 and 11.
- Record rejected alternatives; they are the best defence at audit.

CHAPTER 17

The JM ETA CEO 100

One hundred questions to ask before signing the next SAP contract - organised in twenty domains. Each question is designed to be answerable with a document or a number.

#	Domain	Question
1	SAP Standard	Which standard SAP processes in our current release did we evaluate before approving this change?
2	SAP Standard	Which capabilities are already in our licensed scope but not activated?
3	SAP Standard	Which industry solutions are we licensed for, and which are we actually using?
4	SAP Standard	Is this requirement met in a later feature pack or release we are already entitled to?
5	SAP Standard	Who signed off that standard SAP does not meet this requirement, and on what evidence?
6	Licensing	What do we already own - by product, metric and quantity - and what is its current utilisation?
7	Licensing	Which licences or subscriptions have not been used in the last twelve months?
8	Licensing	How were our users classified, and who challenged the advanced-user count?
9	Licensing	What happens to our commitments if automation reduces transactional activity by 25%?
10	Licensing	Which clause gives us the right to reduce, reclassify or convert licences?
11	RISE	Which components in the RISE bundle do we have a named use case and owner for?
12	RISE	Which Roles and Responsibilities document version is referenced in our contract?
13	RISE	Which tasks that we assume SAP performs are actually chargeable services?
14	RISE	What are our exit terms, data return format and transition costs?
15	RISE	What is the renewal uplift cap, and how was the renewal price modelled?
16	Public / Private / On-Premise	Which processes and entities were assessed for each deployment model?
17	Public / Private / On-Premise	What regulatory or data rules constrain where our ERP can run?
18	Public / Private / On-Premise	How does each model align with our turnaround and shutdown calendar?
19	Public / Private / On-Premise	What is the five- and seven-year TCO of each option on the same basis?
20	Public / Private / On-Premise	Which entities could run on a simpler model without loss of control?
21	AI / Joule	Which Joule capabilities are included in our subscription today, and which are premium?
22	AI / Joule	Which AI use cases have measured benefits from a pilot on our own data?
23	AI / Joule	Which assistants and agents are available for our edition and release?
24	AI / Joule	What prerequisites (BTP, identity, UI technology) must be in place, and at what cost?
25	AI / Joule	Who owns each AI use case and its outcomes?
26	AI Units	What is our contracted price per AI Unit and is it fixed for the term?
27	AI Units	How was our AI Unit quantity forecast, and on what measured consumption?
28	AI Units	What overage rate applies and is it capped?

#	Domain	Question
29	AI Units	How do we monitor consumption monthly, and who is alerted?
30	AI Units	What happens to unused AI Units at the end of each year?
31	BTP	Which BTP services are we consuming, and against which entitlements?
32	BTP	Which business processes stop if BTP or connectivity is unavailable?
33	BTP	Who approves new BTP applications, against which architecture principles?
34	BTP	What is the forecast BTP consumption over five years?
35	BTP	Which BTP applications could run on-stack with released extensibility instead?
36	Clean Core	What is the clean core level distribution of our custom objects today?
37	Clean Core	Which Level D objects are in active use, and what is the plan for each?
38	Clean Core	Who approves exemptions from clean core checks?
39	Clean Core	How will clean core status be measured and reported to the steering committee?
40	Clean Core	What is the cost of clean core remediation versus the upgrade savings it creates?
41	ABAP	How many custom programs exist, and how many were executed in the last twelve months?
42	ABAP	Who owns each business-critical custom program?
43	ABAP	Which custom code exists only to work around poor master data?
44	ABAP	What proportion of upgrade effort is spent on custom code adjustment?
45	ABAP	Have we measured AI-assisted code documentation and testing in our own pilot?
46	Z Tables	What business information do our custom tables hold that standard tables cannot?
47	Z Tables	Which custom tables duplicate data held elsewhere?
48	Z Tables	Which custom tables are read by interfaces, reports or other systems?
49	Z Tables	Which custom tables are safety- or compliance-relevant?
50	Z Tables	Which custom tables will be retired before migration or upgrade?
51	Ariba	Which Ariba modules are we paying for, and what is their usage?
52	Ariba	What percentage of spend goes through catalogues or contracts versus free text?
53	Ariba	Which master data - material, supplier, contract - feeds Ariba, and who owns it?
54	Ariba	Who monitors integration failures between Ariba and ERP, and within what time?
55	Ariba	Are approval limits identical in Ariba and ERP?
56	MM	How many purchase orders are created without a contract or source list?
57	MM	How many purchasing documents are split below approval thresholds?
58	MM	What is our emergency-purchase rate and how is it reviewed?
59	MM	Are price tolerance and three-way match rules consistently applied?
60	MM	How many purchase orders are raised after the invoice?
61	EWM	What warehouse complexity do we have that requires EWM rather than standard inventory management?
62	EWM	Is embedded or decentralised EWM proposed, and why?
63	EWM	What is our current stock accuracy by warehouse?
64	EWM	Which EWM features will we actually use in year one?
65	EWM	What is the incremental licence and support cost of EWM over five years?
66	EAM	What percentage of safety-critical equipment has a validated BOM?

#	Domain	Question
67	EAM	Does our functional-location hierarchy match the physical plant at every site?
68	EAM	What share of work orders are completed with structured damage and cause codes?
69	EAM	Can we trace installed critical components to their certificates in the system?
70	EAM	Which maintenance strategies are based on actual failure data?
71	MDG	Which data domains need central governance workflow, and why?
72	MDG	Who are the named owners and stewards for each domain?
73	MDG	Which data standards exist and are enforced at creation?
74	MDG	What data quality KPIs will prove MDG is working?
75	MDG	Could governance be achieved with existing tools before buying MDG?
76	Master Data	How many duplicate materials and suppliers do we have today?
77	Master Data	What percentage of MRO materials have verified manufacturer and MPN?
78	Master Data	What is our data maturity level per domain and per site?
79	Master Data	What is the budget for data remediation, and is it approved before the technology spend?
80	Master Data	Who is accountable at executive level for master data quality?
81	Integration	How many interfaces do we run, and who owns each one?
82	Integration	Which interfaces are critical during turnarounds and emergencies?
83	Integration	How are failed messages detected and resolved?
84	Integration	Which interfaces exist only because of an overlapping product?
85	Integration	Do our integration and agent designs comply with the SAP API policy version in our contract?
86	Cyber / Security	Which identities - human and non-human - have critical access?
87	Cyber / Security	Are agent identities included in segregation-of-duties analysis?
88	Cyber / Security	What audit logs exist for cloud connectivity and integrations, and who reviews them?
89	Cyber / Security	What are the vendor's incident notification obligations?
90	Cyber / Security	When was our SAP access model last independently tested?
91	Data Sovereignty	Where is each category of our data stored and processed, including backups?
92	Data Sovereignty	Can our data or prompts be used to train vendor models?
93	Data Sovereignty	Which vendor personnel can access our production data?
94	Data Sovereignty	How quickly and in what format can we extract all our data?
95	Data Sovereignty	Which national or sector data rules apply to us, and who confirmed compliance?
96	Commercial / TCO	What is the five-year and seven-year TCO of this decision, including all consumption?
97	Commercial / TCO	What alternatives were costed on the same basis, including "fix the data first"?
98	Commercial / TCO	Which SI effort estimates were independently challenged?
99	Commercial / TCO	Are SI fees tied to activity or to outcomes?
100	Commercial / TCO	What exit strategy is defined, and what does it cost?

CHAPTER 18

25 Sentences That Should Make a CEO Ask for Evidence

None of these sentences is proof of bad faith. Each of them is a signal that a decision is about to be made on assertion rather than evidence.

#	When you hear...	Ask next...
1	"SAP cannot do that."	Which standard process, in which release, was tested? Show the fit-gap and the SAP documentation.
2	"Everyone in the industry buys this add-on."	Which requirement of ours does it meet that our owned scope does not? Name the reference customers we can call.
3	"You need this licence just in case."	Which process will use it, when, and what does it cost if we buy it when we need it instead?
4	"This is the recommended architecture."	Recommended by whom, for which requirements, and what alternatives were rejected and why?
5	"We always implement it this way."	Show why this way fits our processes, not only your delivery method.
6	"This customisation is standard practice."	Standard for whom? What is its clean core level and who will own it?
7	"BTP will solve everything."	For this specific object: consumption forecast, integration pattern, support owner and why on-stack was rejected.
8	"Cloud automatically makes this cheaper."	Show the five- and seven-year TCO for both options on the same basis.
9	"You have to move to Public Cloud to use AI."	Which AI capabilities are available for our edition, release and plan? Show the documentation.
10	"Your Z tables are technical debt."	Which ones, why, what is their usage and clean core level, and what does remediation cost versus retaining them?
11	"You need to buy Joule."	What is included in Base AI with our subscription, and which premium capability, at what AI Unit cost, delivers which measured benefit?
12	"The data will be cleaned during migration."	By whom, to which standard, with what budget, and who sustains it after go-live?
13	"RISE includes everything you need."	Map every bundled component to a use case and every responsibility to a party in the R&R document.
14	"The FUE count comes from our tool."	Show the classification logic; we will run our own analysis before accepting it.
15	"Mainstream maintenance ends - you must decide now."	What are all the options and their costs, including extended maintenance where available?
16	"This price is only valid until quarter end."	What changes in our requirement if we sign next quarter? Our timeline follows evidence, not your quarter.
17	"Our agents are fully autonomous."	Show the approval limits, the audit trail and the stop switch.
18	"Integration is straightforward."	Show the interface inventory, monitoring design and support ownership.
19	"AI will fix the data."	AI can propose corrections. Who approves them, and what standard defines correct?
20	"Upgrades are SAP's responsibility under RISE."	Show the R&R clauses for testing, custom code and functional conversion.
21	"You will lose support if you don't migrate."	What exactly changes in our support, on which date, under which contract clause?
22	"This module is free with the bundle."	Free this term, or at renewal too? What does it cost to operate and integrate?

#	When you hear...	Ask next...
23	"Third-party AI cannot access SAP."	Which policy clause, which version, applied to which of our contracts? Get it in writing.
24	"We need more consultants to de-risk go-live."	Which specific risk, measured how, and why is it not within the existing scope?
25	"Our SI has already validated the requirement."	Validated against whose interest? Who on the owner side signed the requirement independently?

JM ETA ADVISORY PRINCIPLE | The tone to adopt

Ask for evidence calmly and consistently. The goal is not to win an argument with the vendor or the SI; it is to make evidence the normal currency of every SAP decision in your organisation.

CHAPTER 19

The SAP Value Protection Checklist

Seventeen conditions to be evidenced before signing. If a box cannot be ticked, the decision is not yet ready.

	Condition	Evidence	Owner
□	Requirement validated	Signed business requirement	Process owner
□	Standard capability mapped	Fit-gap with SAP references	Solution architect
□	Existing licence mapped	Entitlement and usage report	Licence manager
□	Duplicate capability checked	Capability map across owned products	Enterprise architect
□	Product overlap checked	Overlap analysis	Enterprise architect
□	Architecture validated	Architecture decision record	Design authority
□	Data readiness assessed	Data maturity assessment	Data owner
□	AI readiness assessed	Use-case pilot results and controls	Process owner + risk
□	Integration cost assessed	Interface inventory and cost	Integration lead
□	Customisation cost assessed	Custom code estimate and clean core level	Development lead
□	BTP cost assessed	Consumption forecast	Platform owner
□	AI Unit cost assessed	Rate, forecast, cap, expiry	Finance
□	SI effort challenged	Independent estimate review	Owner-side PMO
□	Five-year TCO calculated	TCO model	Finance
□	Seven-year TCO calculated	TCO model incl. renewal and upgrade	Finance
□	Exit strategy defined	Exit plan and contract clauses	Procurement + legal
□	Governance approved	Steering committee minute	Executive sponsor

EVIDENCE REQUIRED | Signature rule

The executive sponsor should not sign an SAP order form, SI statement of work or change order above the agreed threshold until every item is ticked or a documented, time-bound exception has been approved by the steering committee.

CHAPTER 20

The JM ETA Golden Rules

Fifteen principles distilled from the evidence in this guide.

#	Rule
1	Never buy before mapping what you already own.
2	Never customise before proving standard cannot work.
3	Never accept "SAP cannot do it" without evidence.
4	Never confuse technical capability with licence entitlement.
5	Never confuse cloud with value.
6	Never confuse AI with data readiness.
7	Never automate a broken process.
8	Never automate bad master data.
9	Never sign a long-term commitment without modelling future automation.
10	Never let the SI define the requirement, the solution and the commercial scope without independent challenge.
11	Never accept a responsibility you have not seen written in the Roles and Responsibilities document.
12	Never let an agent approve what a human in the same role could not approve alone.
13	Never let a maintenance date dictate an architecture.
14	Never trade data ownership, portability or exit rights for a discount.
15	Never size a licence from a vendor tool without your own classification analysis.

BEFORE YOU SIGN

If You Are About to Sign - Stop and Ask These 10 Questions

If time allows only one page of this guide to be used, use this one.

1. What exact business problem are we solving, and who in the business signed the requirement?
2. What does standard SAP already provide in our release, and what do we already own?
3. What exact licence, metric, quantity and term are we committing to - and which clause lets us reduce or convert it?
4. What happens to this commitment if automation removes 25% of transactional user activity during the term?
5. What will our AI consumption cost per year, at what fixed rate, with what cap?
6. Which responsibilities stay with us under this contract, according to the Roles and Responsibilities document?
7. Is our master data - materials, suppliers, equipment, hierarchy - ready for what this investment assumes?
8. What is the five-year and seven-year TCO, compared on the same basis with the alternatives?
9. Where will our data live, who can access it, can it train vendor models, and how do we get it back?
10. How do we exit, what does it cost, and who has independently challenged the SI's scope and estimate?

> **NOTE | The governing principle**
>
> SAP is a powerful enterprise platform. The objective is not to buy less SAP for the sake of buying less SAP.
>
> The objective is to buy exactly what the business needs, implement exactly what creates value, retain ownership of the data and architecture, and challenge every additional licence, product, add-on, customisation and consulting cost with evidence.

PROVE THE NEED.
PROVE THE STANDARD.
PROVE THE LICENCE.
PROVE THE ARCHITECTURE.
PROVE THE VALUE.
THEN SPEND.

ABOUT

Jitesh Mehta

Owner-Side. Vendor-Neutral. Evidence-Led.

JM Enterprise Transformation Advisory (JM ETA) is the practice name of Jitesh Mehta. It is an independent, owner-side advisory practice. It has no vendor affiliation, no SI partnership and no agenda other than the client's.

The practice was built on a simple observation: most supply chain and EAM transformations fail not because the technology is wrong, but because the foundation was never built. Master data ungoverned. Asset hierarchies misaligned. Procurement processes that look correct on paper but leave high-risk decisions with a single individual. Vendors and SIs are not positioned to solve this. It is owner-side work.

JM ETA supports boards, CEOs, CSCOs, CIOs, CFOs, COOs, plant managers and transformation leaders across Energy, Oil and Gas, EPC, Power, Utilities and heavy-asset industries, with emphasis on supply chain transformation, EAM transformation, SAP S/4HANA, SAP Ariba, master data, procurement governance, ERP architecture, AI readiness, digital transformation and commercial value protection.

The practice draws on executive-level delivery experience across Shell, ADNOC Group (ZADCO), QatarEnergy, TotalEnergies, Petrofac and Crescent Petroleum.

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APPENDICES

Evidence and Content Engine

The LinkedIn content map derived from this guide, the evidence register behind every verified statement, and the source-document claims that were reviewed and not reproduced as fact.

APPENDIX A

From Guide to Conversation: The Content Engine

Forty-five future LinkedIn posts derived from the chapters of this guide. Each post starts from one chapter and asks one question - it does not repeat the guide.

Ch.	LinkedIn post topic	Executive audience	Core question
1	The seven-step ladder: why BUY is the last word, not the first	CEO, CFO	Which of our last five SAP purchases skipped a step?
1	"SAP cannot do it" is a claim, not a finding	CIO, Transformation Directors	Who tested standard before we paid for custom?
1	A requirement that names a product is a purchase order in disguise	CSCO, Procurement Heads	Who wrote our requirement - the user or the seller?
2	Software deflation is not software getting cheaper	CFO, Board	Which unit of value are we actually paying for?
2	27 software firms named AI agents as a risk. What does that mean for the buyer?	CEO, CIO	Is our contract built for the vendor's risk or ours?
2	Fact, trend or scenario? The three-column test for every AI board paper	Board Members	Is this slide a fact or a forecast?
3	Agents change who operates the ERP - not whether you need one	CIO, CTO	What becomes more valuable when machines do the clicking?
3	The SAP API policy and "bring your own AI"	CIO, Enterprise Architects	Does our agent design comply with the policy in our contract?
3	"You must go to Public Cloud to use AI" - check the documentation	CIO, SAP Leads	Which AI is actually available for our edition?
4	Your automation saving depends on which users disappear	CFO, Licence Managers	Did we automate advanced users or self-service users?
4	The two-meter problem: FUE and AI Units	CFO, CPO	What happens to meter one when meter two grows?
4	One authorisation can promote a user to advanced	CIO, SAP Security	Who challenged our advanced-user count?
4	AI Units expire. Size on evidence, not ambition	CFO	What did we burn versus what did we buy?
4	Technical capability is not licence entitlement	CIO, Legal	Is "already there" also "already paid for"?
5	RISE is not an implementation	CEO, CIO	Which work is still ours after we sign?
5	Read the Roles and Responsibilities document before the order form	CIO, Procurement	Which "included" tasks are chargeable?
5	The exit clause is part of the price	CFO, Legal	What does leaving cost, and when do we find out?
5	Bundled components nobody uses are still paid for	CPO, CFO	Can we name an owner for every bundled item?
6	Public, Private, On-Premise: stop looking for a universal winner	CIO, Board	Which entity fits which model?
6	Public Edition for a subsidiary, Private for the core?	CIO, Group CFO	Are we forcing one model on every entity?
7	A maintenance date is a planning constraint, not a cliff	CEO, CIO	Who turned our support date into a deadline?
7	Who actually does the upgrade work?	COO, Plant Directors	Have we budgeted our own people for testing?

Ch.	LinkedIn post topic	Executive audience	Core question
7	Be careful with upgrade statistics from anyone who sells upgrades	CFO, Board	Where does that percentage come from?
8	Z tables are not automatically technical debt	CIO, SAP Architects	Which custom objects are differentiation and which are debt?
8	SAP's own clean core levels do not ban custom code	SAP Leads, Developers	What level is our code today?
8	Paying an SI to migrate dead code	CFO, PMO	How much of our custom code ran last year?
9	Moving bad logic to BTP does not make it good	CIO, Architects	Why was on-stack extensibility rejected?
9	Every side-by-side extension adds a second lifecycle	CIO, Operations	Which process stops if BTP is down during a shutdown?
10	Ariba does not need a cloud ERP	CPO, CSCO	Were we told otherwise, and by whom?
10	Ariba is a window, not a foundation	CSCO, Category Managers	What percentage of our spend is free text?
10	Two systems, two approval limits, one audit finding	CPO, Internal Audit	Are our limits identical in Ariba and ERP?
11	AI cannot rescue a broken asset foundation	COO, EAM Heads	How many critical assets have a validated BOM?
11	From MTC to installed location: the traceability test	Plant Directors, QA/QC	Can the system prove what is installed?
11	Duplicate materials: stock you own but cannot find	CSCO, Warehouse Managers	What value sits in duplicates?
12	The six levels of data maturity	CEO, CDO	Which level are we really at, by site?
12	MDG is a tool; governance is an organisation	CIO, Data Owners	Who owns our equipment master?
13	The agent that approved its own price variance	CPO, Internal Audit	Who resolves exceptions - the agent or the buyer?
13	Delegation of Authority for digital workers	CFO, Audit Committee	Can an agent approve what a human cannot?
14	One wrong transaction is a problem. A thousand is an enterprise event	Board, CRO	Who can stop the agent, and how fast?
14	Every agent needs a human owner	COO, Department Heads	Whose name is on each agent?
15	The tool is not the moat	CEO, Board	Which data makes us different, and who controls it?
15	Can your data train someone else's model?	CIO, Legal	What do our AI terms say?
16	Twelve questions for every SAP change request	Steering Committees	Which gate did this proposal skip?
18	The sentence "everyone in the industry buys this"	CEO, CPO	Which requirement of ours does it meet?
20	Never trade exit rights for a discount	CFO, Legal	What did the discount really cost?

APPENDIX B

Evidence Register

Every VERIFIED FACT in this guide is traceable to one of the sources below. Sources were accessed in September 2026; SAP pricing, policy and product availability change frequently and should be re-checked before use.

ID	Claim supported	Source type	Location
S01	SAP Business AI packages and pricing: Base AI included in SAP cloud subscriptions; premium AI via AI Units; AI Units purchased annually, expire after 12 months; SAP-delivered autonomous agents 0.02 AI Units per agent action (2026); Joule for Consultants 35 AI Units per user per month; Joule base USD 0.	SAP official product/pricing page	sap.com/products/artificial-intelligence/pricing.html
S03	FUE ratios: 1 FUE = 1 advanced use, 5 core use, 30 self-service use; developer access 2 FUE.	SAP Community (SAP-authored blog)	community.sap.com - "RISE with SAP Full Use Equivalent FUE concept"
S04	SAP publishes no FUE list price; ratios fixed; no mid-term reduction; over-classification common.	Independent licensing adviser (Redress Compliance, Aug 2026)	redresscompliance.com/sap-fue-licensing-explained...
S05	SAP 2023 cloud-only innovation statement; Sapphire May 2026: majority of Joule assistants/agents for on-prem S/4HANA and ECC customers committed to RISE journey, tied to Max Success Plan.	Reputable trade press (The Register, 13 May 2026)	theregister.com/ai-ml/2026/05/13/sap-u-turn-brings-ai-features-to-ecc-and-on-prem-s/4hana/5239040
S06	SAP knowledge base article titled "Joule is not available for On-Premise systems" (KBA 3632703).	SAP Support (KBA title, preview)	userapps.support.sap.com/sap/support/knowledge/en/3632703
S07	Joule for Developers ABAP AI offered for BTP ABAP Environment, Public Edition and Private Edition 2021+; on-premise 2021-2025 subject to special RISE contract conditions.	SAP Community (June 2026)	community.sap.com - "SAP Joule for Developers, ABAP AI capabilities..."
S08	Joule available for S/4HANA Cloud Private Edition from release 2021; new RISE contracts auto-entitled.	SAP Community (May 2026)	community.sap.com - "SAP Joule for SAP S/4HANA Cloud Private Edition (PCE) - Detailed steps"
S09	SAP API Policy v.4/2026 Section 2.2.2: restricts API use with (semi-)autonomous or generative AI systems that plan, select or execute sequences of API calls, except via SAP-endorsed architectures.	SAP Community member analysis of policy text	community.sap.com - "SAP's API Policy on AI Agents: What It Prohibits, Allows, and Leaves Open"
S10	API policy enforcement from 9 June 2026 via security patch affecting ODP via RFC.	Independent analyst (Forrester blog, May 2026)	forrester.com/blogs/sap-is-attempting-to-become-the-gatekeeper-of-enterprise-ai...
S11	SAP CEO clarified intent: protect domain know-how and prevent performance degradation; not to block customers from their own data.	Independent commentary (Kai Waehner, 2026)	kai-waehner.de/blog/2026/05/02/data-ownership-in-the-age-of-agentic-ai...
S12	From S/4HANA 2023: two-year release cycle, seven years mainstream maintenance, feature packs every six months; at least one release in maintenance until 2040.	SAP News Center (Sept 2022)	news.sap.com/2022/09/new-sap-s4hana-release-maintenance-strategy/
S13	New S/4HANA release every two years; definitions of upgrade vs update; SAP Note 2900388.	SAP Help Portal	help.sap.com - Release Cycles in SAP S/4HANA
S14	Private Edition customers not forced to upgrade within mainstream maintenance; Public Edition upgraded twice a year, mandatory.	SAP Community Q&A (SAP response)	groups.community.sap.com - release cycle and forced version up of S/4HANA Cloud private edition
S15	After mainstream maintenance ends, release moves to customer-specific maintenance for customers with valid support.	SAP partner summary of SAP maintenance concepts (Jan 2026)	soapeople.com/blog/end-of-support-for-sap-ecc-and-sap-s/4hana
S16	ERP private edition transition option 2031-2033 for large, complex RISE customers; not a prolongation of 2027/2030 deadlines.	Independent analyst (Forrester blog)	forrester.com/blogs/sap-offers-on-prem-erp-customers-additional-3-year-support...

ID	Claim supported	Source type	Location
S17	ECC EHP 6-8 mainstream maintenance ends 2027; extended maintenance to end-2030 at +2 percentage points.	SAP partner (IBsolution, May 2026)	ibsolution.com/academy/blog_en/.../end-of-maintenance-for-sap-ecc-in-2027...
S18	RISE Roles & Responsibilities: application problems resolved by customer; Cloud Application Services at additional charge; overall system conversion responsibility with customer; no OS access for customers.	SAP official R&R documents (v7-2022; tailored option v3-2025)	assets.cdn.sap.com/agreements/product-policy/hec/roles-responsibilities/...
S19	RISE Private Edition on AWS: single-tenant; SAP owns and manages the AWS account; SAP responsible for holistic SLA, cloud operations, technical support and security in the cloud; bundle includes BPI, platform/analytics, business networks.	AWS official documentation	docs.aws.amazon.com/sap/latest/general/rise.html
S20	Clean core level concept A-D (2025): Level A on-stack ABAP Cloud or side-by-side BTP using released interfaces; Level B classic APIs considered clean; C conditionally clean; D not clean core.	SAP News Center (Aug 2025) and SAP Community (SAP-authored)	news.sap.com/2025/08/extend-sap-s4hana-cloud-right-way-clean-clear/
S21	Managed gateway for spend management and SAP Business Network (formerly Cloud Integration Gateway) integrates SAP ERP and S/4HANA with SAP Ariba and Business Network; add-on for ERP and S/4HANA.	SAP Help Portal	help.sap.com/docs/sisgw/...
S22	SAP Cloud Connector: on-premise agent, reverse-invoke proxy, no inbound firewall ports, access control, audit logging.	SAP Learning	learning.sap.com - Explaining the SAP Cloud Connector
S23	27 software companies listed AI agents as a competitive risk in filings in early 2026 vs 7 a year earlier (The Information / AlphaSense analysis).	Secondary report of The Information analysis (March 2026)	The Information; summary via itiger.com news 1174623114
S24	Anthropic preview agent (Jan 2026) able to take over mouse and keyboard to work across apps; OpenAI tool for running many agents across software applications.	The Information ("Agent Toll Gates", Feb 2026)	theinformation.com/articles/agent-toll-gates-software-companies-ponder-respond-ai-risks
S25	SAP CEO: SAP will charge based on AI consumption, moving away from regular subscriptions (March 2026).	Bloomberg (18 Mar 2026)	bloomberg.com/news/articles/2026-03-18/sap-ceo-pushes-ai-turnaround-with-new-teams-use-based-pricing
S26	SAP marketing claim: Utilities Customer Self-Service Agent can cut customer service costs "by up to 90%*" (footnoted vendor claim).	SAP industry web page	sap.com/uk/industries/utilities.html
S27	Marketing names: S/4HANA Cloud Public Edition = SAP Cloud ERP; Private Edition = SAP Cloud ERP Private; licensed names unchanged.	SAP partner (TIMETOACT)	timetoact-group.com/.../sap-simplifies-the-naming-of-its-cloud-erp-offering
S28	Private Edition single-tenant; Public Edition multi-tenant with shared infrastructure.	SAP partner (Gambit)	gambit.de/en/wiki/sap-private-cloud
S30	Reported SAP Learning consumption tiers: basic/standard/advanced agents 0.005/0.01/0.025 AI Units per step; Joule for Consultants pooled 22,900 requests.	Secondary summary of SAP Learning (Inclusion Cloud, Jul 2026)	inclusioncloud.com/insights/control-joule-ai-unit-spending-sap/

Source IDs S02 and S29 are intentionally unused. Independent advisory sources (S04, S10, S11, S30) are cited as independent observations, not as SAP statements.

APPENDIX C

Source-Document Claims Reviewed

The primary source document contained claims that were outdated, unverified or opinion. They are listed here so that readers can see exactly what was corrected or excluded, and why.

Claim in source document	Status	Finding
On-Premise S/4HANA cannot get SAP AI or Joule agents at all.	Outdated / partially incorrect	SAP announced in May 2026 availability of most Joule assistants and agents for on-prem S/4HANA and ECC customers committed to the RISE journey under a specific plan [S05]. Eligibility is contract-specific.
Custom OpenAI/Azure agents over on-prem SAP let the owner capture 100% of automation savings.	Not established	SAP API Policy v.4/2026 restricts third-party autonomous AI access to SAP APIs outside endorsed architectures [S09-S11]. Costs of building, securing and supporting agents are also not zero.
"Joule for Developers" costs 35 AI Units per user per month.	Corrected	SAP's pricing page attributes 35 AI Units per user per month to SAP Joule for Consultants [S01].
Private Edition receives a major upgrade once a year.	Outdated	Since S/4HANA 2023, releases are every two years with feature packs [S12, S13].
Extended maintenance costs 2% to 4% extra.	Corrected	Published ECC extended maintenance premium is two percentage points [S16, S17]; other figures require contract validation.
Over 60% of early S/4HANA adopters did not upgrade; 15-20% of heavy-asset firms move to third-party support; 80% choose purely technical upgrades.	Not verified	No primary, transparent source located. Excluded from the guide.
AI agent vs human PO metrics: 45-90 seconds vs 12-20 minutes; error rate below 0.5% vs 5-8%.	Not verified	Illustrative claims without source. Excluded; measure in own pilot.
AI coding tools cut ABAP maintenance costs by 60-70%.	Not verified	Plausible direction, unproven magnitude. Measure in own pilot.
Public Edition is useless for the Energy industry.	Opinion	Too absolute. Public Edition may fit subsidiaries or standard entities; fit-gap decides.
Private Edition is essentially the old on-premise system in a subscription wrapper.	Opinion	Partly true for application scope; incomplete - SAP operates infrastructure and technical operations [S19].
SAP retired per-user Joule packages; all SAP AI is consumption-only.	Partially incorrect	SAP prices premium AI either per user per month or by consumption depending on the feature [S01].
Under RISE, FUE cannot be reduced mid-contract.	Consistent with independent reporting	Reported by independent advisers [S04]; confirm in your own order form.
27 public software firms flagged AI agents as a material risk.	Verified via secondary report	The Information / AlphaSense analysis reported 27 vs 7 [S23]; full list is paywalled.
SAP CEO said charging subscriptions would be "foolish".	Direction verified; exact wording not verified	Bloomberg reports SAP will move to AI consumption-based charging [S25]; exact quotation not independently confirmed.
SAP Utilities self-service agent reduces service costs by up to 90%.	Vendor claim	Appears as a footnoted SAP marketing claim [S26]; treat as vendor claim, not an outcome.
Ariba with on-prem ERP is the most common heavy-industry architecture.	Not verified	Hybrid coexistence is documented [S21]; prevalence is not.